# L2 Geometry Workbook for Profile Sieve EL

Three-Part Proof: Geometry, Small Stochastic Bounds, and Merging

Solved derivation workbook

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# Part I

# Geometric and Convex KKT Body

## 1 Model Setup: Full Geometry in $L^2(P_T)$ and $L^2(P)$

The purpose of this workbook is to rewrite the whole proof logic as Hilbert space geometry. The sample space is not first probability; it is first the finite empirical Hilbert space

$$L^2(P_T) = (\mathbb{R}^T, \langle a, b\rangle_T = T^{-1}a'b).$$

The population limit is the Hilbert space

$$L^2(P), \qquad \langle f, g\rangle_P = \mathbb{E}\{fg\}.$$

The model setup is the following chain:

$$\begin{aligned}
\boldsymbol{y} &= \beta_0\boldsymbol{x} + \boldsymbol{\mu}_0 + \boldsymbol{u} && \text{model in } L^2(P_T),\\
\mu_{0,t} &= m_{j_0(t)}(W_{t-1}), \quad j_0(t) = j \iff k_{j,0} < t \le k_{j+1,0} && \text{true nuisance vector},\\
\mathcal{N}_{K,\Lambda,T} &= \mathrm{col}(\boldsymbol{Q}_{K,\Lambda}) && \text{nuisance subspace},\\
\boldsymbol{M}_{K,\Lambda} &= I - \Pi_{K,\Lambda,T} && \text{residual-maker},\\
S_T(\beta,\Lambda) &= \sqrt{T}\langle \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y}-\beta\boldsymbol{x}), \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\rangle_T && \text{projected score}.
\end{aligned}$$

Here $\boldsymbol{\mu}_0$ is exactly the stacked nonparametric nuisance part of the model. If there are no nuisance breaks, $\mu_{0,t} = m(W_{t-1})$. With breaks, $\mu_{0,t} = m_j(W_{t-1})$ on regime $j$, so $\boldsymbol{\mu}_0$ is not one stable function evaluated over the whole sample; it is the true piecewise nuisance surface evaluated observation by observation.

EL then uses the scalar products

$$Z_t(\beta,\Lambda) = \{\boldsymbol{M}_{K,\Lambda}(\boldsymbol{y}-\beta\boldsymbol{x})\}_t\{\boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\}_t$$

to form a one-dimensional empirical likelihood. EL is not the Hilbert projection; EL is the likelihood wrapper around the projected score.

**Three-part proof map.** Part I is finite-sample geometry: profiling, residual scores, and EL are all KKT/convex objects in $L^2(P_T)$. Part II is small stochastic verification: Gram stability, sieve and break errors, CLT, variance, max-score, and convex-hull size. Part III merges them: consistency comes from the moment bridge plus unique population zero, while Wilks comes from the EL KKT quadratic reduction plus the standardized score CLT.

# 2 Finite-Sample Hilbert Space: Viewing the Entire Model in $L^2(P_T)$

## 2.1 The empirical space

Let

$$\Omega_T = \{1, \ldots, T\}, \qquad P_T = T^{-1} \sum_{t=1}^{T} \delta_t.$$

Every sample vector $\boldsymbol{a} = (a_1, \ldots, a_T)'$ is a function $a_T : \Omega_T \to \mathbb{R}$, with $a_T(t) = a_t$. The empirical inner product is

$$\langle \boldsymbol{a}, \boldsymbol{b} \rangle_T = \int a_T b_T \, dP_T = T^{-1} \sum_{t=1}^{T} a_t b_t = T^{-1} \boldsymbol{a}' \boldsymbol{b}.$$

Thus all variables in the sample are elements of $L^2(P_T)$:

$$\boldsymbol{y}, \boldsymbol{x}, \boldsymbol{w}, \boldsymbol{u}, \boldsymbol{\mu}_0 \in L^2(P_T).$$

## 2.2 The full finite-sample model

The break-aware model is the vector identity in $L^2(P_T)$

$$\boldsymbol{y} = \beta_0 \boldsymbol{x} + \boldsymbol{\mu}_0 + \boldsymbol{u}, \tag{2.1}$$

where

$$\mu_{0,t} = \sum_{j=0}^{q_0} m_j(W_{t-1}) \mathbf{1}\{k_{j,0} < t \leq k_{j+1,0}\}.$$

So the full model says: $\boldsymbol{y}$ lies in the affine set

$$\beta_0 \boldsymbol{x} + \boldsymbol{\mu}_0 + \boldsymbol{u}.$$

The inferential problem is to remove $\boldsymbol{\mu}_0$, not by estimating it as a main parameter, but by projecting away its nuisance subspace.

## 2.3 Candidate nuisance subspaces

For a candidate partition $\Lambda = (k_1, \ldots, k_q)$, define the block sieve matrix

$$\boldsymbol{Q}_{K,\Lambda} = [\boldsymbol{D}_0(\Lambda)\boldsymbol{P}, \ldots, \boldsymbol{D}_q(\Lambda)\boldsymbol{P}].$$

The sample nuisance subspace is

$$\mathcal{N}_{K,\Lambda,T} = \operatorname{col}(\boldsymbol{Q}_{K,\Lambda}) \subset L^2(P_T). \tag{2.2}$$

An element of this space has the form

$$g_T(t) = \sum_{j=0}^{q} \mathbf{1}\{t \in I_j(\Lambda)\} \boldsymbol{P}_K(W_{t-1})' c_j.$$

This is the finite-sample version of a regime-specific nonparametric nuisance function.

Let $j_\Lambda(t)$ denote the regime index induced by $\Lambda$. For a regime-wise nuisance tuple $h = (h_0, \ldots, h_q)$, define its stacked sample vector by

$$\boldsymbol{h}_{\Lambda,T} = \big(h_{j_\Lambda(1)}(W_0), \ldots, h_{j_\Lambda(T)}(W_{T-1})\big)'.$$

This vector is the empirical version of the candidate nuisance direction.

**Lemma 1** (Uniform finite-sieve approximation in $L^2(P_T)$)**.** *Fix $(K, \Lambda, T)$ and a nuisance class $\mathcal{H}$. For each $h \in \mathcal{H}$, let $\boldsymbol{h}_{\Lambda,T} \in L^2(P_T)$ be the stacked nuisance vector defined above. Define the finite-sieve approximation radius*

$$\rho_{K,T}(\mathcal{H}, \Lambda) = \sup_{h \in \mathcal{H}} \inf_{\gamma \in \mathbb{R}^{(q+1)K}} \|\boldsymbol{h}_{\Lambda,T} - \boldsymbol{Q}_{K,\Lambda}\gamma\|_T. \tag{2.3}$$

*If $\rho_{K,T}(\mathcal{H}, \Lambda) = o_p(1)$, then the desired uniform $L^2(P_T)$ approximation holds:*

$$\sup_{h \in \mathcal{H}} \|\boldsymbol{h}_{\Lambda,T} - \boldsymbol{Q}_{K,\Lambda}\gamma_{K,\Lambda}(h)\|_T = \sup_{h \in \mathcal{H}} \|\boldsymbol{r}_{K,\Lambda,T}(h)\|_T = \rho_{K,T}(\mathcal{H}, \Lambda) = o_p(1). \tag{2.4}$$

*Here, for each $h$,*

$$\gamma_{K,\Lambda}(h) \in \operatorname*{argmin}_{\gamma \in \mathbb{R}^{(q+1)K}} \|\boldsymbol{h}_{\Lambda,T} - \boldsymbol{Q}_{K,\Lambda}\gamma\|_T^2,$$

$$\boldsymbol{h}_{K,\Lambda,T} = \boldsymbol{Q}_{K,\Lambda}\gamma_{K,\Lambda}(h), \qquad \boldsymbol{r}_{K,\Lambda,T}(h) = \boldsymbol{h}_{\Lambda,T} - \boldsymbol{h}_{K,\Lambda,T}.$$

*The ambient space is $L^2(P_T) \cong \mathbb{R}^T$, so $\boldsymbol{h}_{\Lambda,T}$, $\boldsymbol{h}_{K,\Lambda,T}$, and $\boldsymbol{r}_{K,\Lambda,T}(h)$ are all $T \times 1$ vectors. The design matrix and coefficient vector have dimensions*

$$\boldsymbol{Q}_{K,\Lambda} \in \mathbb{R}^{T \times (q+1)K}, \qquad \gamma_{K,\Lambda}(h) \in \mathbb{R}^{(q+1)K}.$$

*The finite nuisance space has dimension*

$$d_{K,\Lambda,T} = \operatorname{rank}(\boldsymbol{Q}_{K,\Lambda}) \leq (q+1)K,$$

*and*

$$\boldsymbol{h}_{\Lambda,T} = \boldsymbol{Q}_{K,\Lambda}\gamma_{K,\Lambda}(h) + \boldsymbol{r}_{K,\Lambda,T}(h), \tag{2.5}$$

*where*

$$\boldsymbol{Q}_{K,\Lambda}\gamma_{K,\Lambda}(h) \in \mathcal{N}_{K,\Lambda,T} \subset L^2(P_T), \qquad \boldsymbol{r}_{K,\Lambda,T}(h) \in \mathcal{N}_{K,\Lambda,T}^{\perp_T} \subset L^2(P_T).$$

*Thus $\boldsymbol{r}_{K,\Lambda,T}(h)$ is a $T \times 1$ approximation-residual vector living in a $(T - d_{K,\Lambda,T})$-dimensional orthogonal complement, and*

$$\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{r}_{K,\Lambda,T}(h) = 0.$$

*Proof.* The column space $\mathcal{N}_{K,\Lambda,T} = \text{col}(\boldsymbol{Q}_{K,\Lambda})$ is a finite-dimensional closed subspace of $L^2(P_T)$. Therefore the Hilbert projection theorem gives the closest finite-sieve approximation to every $\boldsymbol{h}_{\Lambda,T}$:

$$\Pi_{K,\Lambda,T}\boldsymbol{h}_{\Lambda,T} \in \mathcal{N}_{K,\Lambda,T}.$$

Because every element of $\mathcal{N}_{K,\Lambda,T}$ is $\boldsymbol{Q}_{K,\Lambda}\gamma$, the projection can be written as

$$\Pi_{K,\Lambda,T}\boldsymbol{h}_{\Lambda,T} = \boldsymbol{Q}_{K,\Lambda}\gamma_{K,\Lambda}(h)$$

for any least-squares minimizer $\gamma_{K,\Lambda}(h)$. If $\boldsymbol{Q}_{K,\Lambda}$ is rank deficient, the coefficient vector need not be unique, but the fitted vector is unique.

Define

$$\boldsymbol{r}_{K,\Lambda,T}(h) = \boldsymbol{h}_{\Lambda,T} - \Pi_{K,\Lambda,T}\boldsymbol{h}_{\Lambda,T}.$$

The projection theorem gives

$$\langle \boldsymbol{r}_{K,\Lambda,T}(h), \boldsymbol{v}\rangle_T = 0 \qquad \text{for every } \boldsymbol{v} \in \mathcal{N}_{K,\Lambda,T},$$

which implies $\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{r}_{K,\Lambda,T}(h) = 0$. This proves the orthogonal decomposition (2.5) and the location $\boldsymbol{r}_{K,\Lambda,T}(h) \in \mathcal{N}^{\perp_T}_{K,\Lambda,T}$.

By the definition of orthogonal projection,

$$\|\boldsymbol{r}_{K,\Lambda,T}(h)\|_T = \inf_{\gamma\in\mathbb{R}^{(q+1)K}} \|\boldsymbol{h}_{\Lambda,T} - \boldsymbol{Q}_{K,\Lambda}\gamma\|_T.$$

Taking the supremum over $h \in \mathcal{H}$ gives exactly (2.4). Hence the mathematical task behind this lemma is to verify $\rho_{K,T}(\mathcal{H},\Lambda) = o_p(1)$ from the chosen basis, smoothness class, and growth rate of $K$. An arbitrary nuisance direction is not assumed to be exactly spanned by the sieve basis; it is approximated by its finite-sieve projection, and the leftover is $\boldsymbol{r}_{K,\Lambda,T}(h)$. □

## 2.4 Orthogonal projection

Let $\Pi_{K,\Lambda,T}$ be the $L^2(P_T)$-orthogonal projection onto $\mathcal{N}_{K,\Lambda,T}$. In coordinates,

$$\Pi_{K,\Lambda,T} = \boldsymbol{Q}_{K,\Lambda}(\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda})^{\dagger}\boldsymbol{Q}'_{K,\Lambda}.$$

The residual-maker is

$$\boldsymbol{M}_{K,\Lambda} = I - \Pi_{K,\Lambda,T}. \tag{2.6}$$

It satisfies the projection identities

$$\boldsymbol{M}'_{K,\Lambda} = \boldsymbol{M}_{K,\Lambda}, \qquad \boldsymbol{M}^2_{K,\Lambda} = \boldsymbol{M}_{K,\Lambda}, \qquad \boldsymbol{Q}'_{K,\Lambda}\boldsymbol{M}_{K,\Lambda} = 0.$$

**Lemma 2** (Projection identities for the residual-maker)**.** *Let $Q = \boldsymbol{Q}_{K,\Lambda}$, $G = Q'Q$,*

$$\Pi = QG^{\dagger}Q', \qquad M = I - \Pi,$$

*where $G^{\dagger}$ is the Moore-Penrose inverse. Then $\Pi$ and $M$ are symmetric and idempotent, and $M$ annihilates the sieve nuisance space:*

$$M' = M, \qquad M^2 = M, \qquad MQ = 0, \qquad Q'M = 0.$$

*Proof.* Since $G = Q'Q$ is symmetric, its Moore-Penrose inverse $G^\dagger$ is also symmetric. Hence

$$\Pi' = \{QG^\dagger Q'\}' = QG^\dagger Q' = \Pi.$$

For idempotence, use the Moore-Penrose identity $G^\dagger G G^\dagger = G^\dagger$:

$$\Pi^2 = QG^\dagger Q'QG^\dagger Q' = QG^\dagger G G^\dagger Q' = QG^\dagger Q' = \Pi.$$

Therefore

$$M' = I - \Pi' = M, \qquad M^2 = (I - \Pi)^2 = I - 2\Pi + \Pi^2 = I - \Pi = M.$$

It remains to show that $M$ kills the columns of $Q$. Since $\mathcal{R}(G) = \mathcal{R}(Q')$ and $G^\dagger G$ is the orthogonal projector onto $\mathcal{R}(G)$,

$$QG^\dagger G = Q.$$

Thus

$$\Pi Q = QG^\dagger Q'Q = QG^\dagger G = Q, \qquad MQ = (I - \Pi)Q = 0.$$

Finally, because $M$ is symmetric, $Q'M = (MQ)' = 0$. This proves all projection identities. $\square$

# 3 Profiling Is Orthogonal Projection

For a trial value $\beta$, define the partial outcome

$$\boldsymbol{y}^\circ_\beta = \boldsymbol{y} - \beta\boldsymbol{x}.$$

It is the original outcome $\boldsymbol{y}$ after subtracting the trial linear component $\beta\boldsymbol{x}$.

Profiling the nuisance means finding the closest nuisance vector to the partial outcome $\boldsymbol{y}^\circ_\beta$ in $\mathcal{N}_{K,\Lambda,T}$:

$$\widehat{\boldsymbol{\mu}}_{\beta,\Lambda} = \arg\min_{\boldsymbol{g}\in\mathcal{N}_{K,\Lambda,T}} \|\boldsymbol{y}^\circ_\beta - \boldsymbol{g}\|^2_T.$$

By Hilbert projection,

$$\widehat{\boldsymbol{\mu}}_{\beta,\Lambda} = \Pi_{K,\Lambda,T}\boldsymbol{y}^\circ_\beta, \qquad \widehat{\boldsymbol{u}}(\beta,\Lambda) = \boldsymbol{y}^\circ_\beta - \widehat{\boldsymbol{\mu}}_{\beta,\Lambda} = \boldsymbol{M}_{K,\Lambda}\boldsymbol{y}^\circ_\beta.$$

**Lemma 3** (Profile normal equation)**.** *For every $\boldsymbol{h} \in \mathcal{N}_{K,\Lambda,T}$,*

$$\langle\widehat{\boldsymbol{u}}(\beta,\Lambda), \boldsymbol{h}\rangle_T = 0.$$

*Equivalently,*

$$\boldsymbol{Q}'_{K,\Lambda}\widehat{\boldsymbol{u}}(\beta,\Lambda) = 0.$$

*Proof.* This is the KKT condition for a least-squares projection in $L^2(P_T)$. If $\boldsymbol{h} = \boldsymbol{Q}_{K,\Lambda}\boldsymbol{a}$, then

$$\langle\widehat{\boldsymbol{u}}, \boldsymbol{h}\rangle_T = T^{-1}\widehat{\boldsymbol{u}}'\boldsymbol{Q}_{K,\Lambda}\boldsymbol{a} = T^{-1}(\boldsymbol{Q}'_{K,\Lambda}\widehat{\boldsymbol{u}})'\boldsymbol{a} = 0.$$

$\square$

## 3.1 Break search as subspace selection

For fixed $\beta$, the partial outcome is

$$\boldsymbol{y}_\beta^\circ = \boldsymbol{y} - \beta \boldsymbol{x}.$$

For a candidate partition $\Lambda = (k_1, \dots, k_q)$, set $k_0 = 0$, $k_{q+1} = T$, and

$$I_j(\Lambda) = \{t : k_j < t \le k_{j+1}\}.$$

The block sieve matrix is

$$\boldsymbol{Q}_{K,\Lambda} = [\boldsymbol{D}_0(\Lambda)\boldsymbol{P}, \dots, \boldsymbol{D}_q(\Lambda)\boldsymbol{P}].$$

For $c = (c_0', \dots, c_q')'$, the fitted nuisance vector is $\boldsymbol{Q}_{K,\Lambda}c$, with component

$$(\boldsymbol{Q}_{K,\Lambda}c)_t = \boldsymbol{P}_K(W_{t-1})'c_j \qquad \text{if } t \in I_j(\Lambda).$$

Thus minimizing over $c$ is exactly the same as choosing a separate sieve coefficient vector $c_j$ on each segment.

**Lemma 4** (Profile RSS is a Hilbert distance)**.** *For fixed $(\beta, \Lambda)$, define the segment cost*

$$C_K(s, e; \beta) = \min_{c \in \mathbb{R}^K} \sum_{t=s}^{e} \{y_t - \beta x_t - \boldsymbol{P}_K(W_{t-1})'c\}^2,$$

*and the profile RSS*

$$RSS_K(\beta, \Lambda) = \sum_{j=0}^{q} C_K(k_j + 1, k_{j+1}; \beta).$$

*Then*

$$\begin{aligned} RSS_K(\beta, \Lambda) &= \min_{c \in \mathbb{R}^{(q+1)K}} \|\boldsymbol{y}_\beta^\circ - \boldsymbol{Q}_{K,\Lambda}c\|_2^2 \\ &= \inf_{\boldsymbol{g} \in \mathcal{N}_{K,\Lambda,T}} \|\boldsymbol{y}_\beta^\circ - \boldsymbol{g}\|_2^2 \\ &= T\, \mathrm{dist}_T^2(\boldsymbol{y}_\beta^\circ, \mathcal{N}_{K,\Lambda,T}) \\ &= T\|\boldsymbol{M}_{K,\Lambda}\boldsymbol{y}_\beta^\circ\|_T^2. \end{aligned} \tag{3.1}$$

*Proof.* Because each segment $I_j(\Lambda)$ has its own coefficient vector $c_j$,

$$\begin{aligned} RSS_K(\beta, \Lambda) &= \min_{c_0, \dots, c_q} \sum_{j=0}^{q} \sum_{t \in I_j(\Lambda)} \{y_t - \beta x_t - \boldsymbol{P}_K(W_{t-1})'c_j\}^2 \\ &= \min_{c \in \mathbb{R}^{(q+1)K}} \|\boldsymbol{y} - \beta\boldsymbol{x} - \boldsymbol{Q}_{K,\Lambda}c\|_2^2 \\ &= \min_{c \in \mathbb{R}^{(q+1)K}} \|\boldsymbol{y}_\beta^\circ - \boldsymbol{Q}_{K,\Lambda}c\|_2^2. \end{aligned}$$

Since $\mathcal{N}_{K,\Lambda,T} = \mathrm{col}(\boldsymbol{Q}_{K,\Lambda})$, every $\boldsymbol{g} \in \mathcal{N}_{K,\Lambda,T}$ can be written as $\boldsymbol{g} = \boldsymbol{Q}_{K,\Lambda}c$. Therefore

$$\min_c \|\boldsymbol{y}_\beta^\circ - \boldsymbol{Q}_{K,\Lambda}c\|_2^2 = \inf_{\boldsymbol{g} \in \mathcal{N}_{K,\Lambda,T}} \|\boldsymbol{y}_\beta^\circ - \boldsymbol{g}\|_2^2.$$

The Euclidean norm and the empirical Hilbert norm satisfy $\|\boldsymbol{a}\|_2^2 = T\|\boldsymbol{a}\|_T^2$. Hence

$$\inf_{\boldsymbol{g}\in\mathcal{N}_{K,\Lambda,T}} \|\boldsymbol{y}_\beta^\circ - \boldsymbol{g}\|_2^2 = T\,\mathrm{dist}_T^2(\boldsymbol{y}_\beta^\circ, \mathcal{N}_{K,\Lambda,T}).$$

Finally, $\boldsymbol{M}_{K,\Lambda}$ is the residual-maker for the orthogonal projection onto $\mathcal{N}_{K,\Lambda,T}$, so

$$\mathrm{dist}_T^2(\boldsymbol{y}_\beta^\circ, \mathcal{N}_{K,\Lambda,T}) = \|\boldsymbol{M}_{K,\Lambda}\boldsymbol{y}_\beta^\circ\|_T^2.$$

This proves (3.1). □

Consequently, for known order $q$, the break estimator is

$$\begin{aligned}\widehat{\Lambda}_q(\beta) &= \arg\min_{\Lambda\in\mathcal{L}_q(\epsilon)} RSS_K(\beta,\Lambda)\\ &= \arg\min_{\Lambda\in\mathcal{L}_q(\epsilon)} \mathrm{dist}_T^2(\boldsymbol{y}_\beta^\circ, \mathcal{N}_{K,\Lambda,T}).\end{aligned}$$

Unknown order adds a penalty to the same distance:

$$\widehat{q}(\beta) = \arg\min_{0\le q\le \bar{q}} \left\{T\,\mathrm{dist}_T^2(\boldsymbol{y}_\beta^\circ, \mathcal{N}_{K,\widehat{\Lambda}_q(\beta),T}) + \mathrm{pen}_T(q,K)\right\}.$$

Thus break search is subspace selection: each candidate $\Lambda$ creates a candidate nuisance subspace $\mathcal{N}_{K,\Lambda,T}$, and RSS picks the subspace closest to the partial outcome $\boldsymbol{y}_\beta^\circ$.

# 4 Residualized Score Direction

The raw score weight $\boldsymbol{w} \in L^2(P_T)$ may contain nuisance directions. The geometric score direction is its nuisance-orthogonal component

$$\boldsymbol{v}_\Lambda = \boldsymbol{w}^c(\Lambda) = \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}.$$

Then

$$\boldsymbol{v}_\Lambda \in \mathcal{N}_{K,\Lambda,T}^{\perp_T}, \qquad \langle \boldsymbol{v}_\Lambda, \boldsymbol{h}\rangle_T = 0 \quad \forall \boldsymbol{h}\in\mathcal{N}_{K,\Lambda,T}.$$

The sample profiled moment is the Hilbert inner product

$$\Psi_T(\beta,\Lambda) = \langle \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y}-\beta\boldsymbol{x}), \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\rangle_T. \tag{4.1}$$

The scaled score is

$$S_T(\beta,\Lambda) = \sqrt{T}\,\Psi_T(\beta,\Lambda).$$

**Lemma 5** (Finite-sample nuisance cancellation)**.** *If* $\boldsymbol{g}\in\mathcal{N}_{K,\Lambda,T}$, *then*

$$\langle \boldsymbol{M}_{K,\Lambda}\boldsymbol{g}, \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\rangle_T = 0.$$

*Proof.* Because $\boldsymbol{g}\in\mathcal{N}_{K,\Lambda,T}$, $\boldsymbol{M}_{K,\Lambda}\boldsymbol{g} = 0$. Hence the inner product is zero. Equivalently, $(\boldsymbol{Q}c)'\boldsymbol{M}_{K,\Lambda}\boldsymbol{w} = 0$ for every $c$. □

# 5 Known True Partition: Wilks Logic in Geometry

Set $\Lambda = \Lambda_0$, and abbreviate

$$\boldsymbol{Q}_0 = \boldsymbol{Q}_{K,\Lambda_0}, \qquad \boldsymbol{M}_0 = \boldsymbol{M}_{K,\Lambda_0}, \qquad \boldsymbol{v}_0 = \boldsymbol{M}_0\boldsymbol{w}.$$

Decompose the true nuisance into its finite-sieve projection and approximation residual:

$$\boldsymbol{\mu}_0 = \boldsymbol{Q}_0\boldsymbol{c}_0 + \boldsymbol{r}_0, \qquad \boldsymbol{Q}_0\boldsymbol{c}_0 \in \mathcal{N}_{K,\Lambda_0,T}, \qquad \boldsymbol{r}_0 \in \mathcal{N}_{K,\Lambda_0,T}^{\perp_T}.$$

**Lemma 6** (True-partition score decomposition)**.** *At $\beta = \beta_0$ and $\Lambda = \Lambda_0$, the projected score satisfies*

$$S_T(\beta_0, \Lambda_0) = T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0 + T^{-1/2}\boldsymbol{r}_0'\boldsymbol{v}_0. \tag{5.1}$$

*The first term is the stochastic score. The second term is the sieve approximation-bias term, and this is the term controlled by approximation rates:*

$$T^{-1/2}\boldsymbol{r}_0'\boldsymbol{v}_0 = T^{-1/2}\boldsymbol{r}_0'\boldsymbol{M}_0\boldsymbol{w} = o_p(1). \tag{5.2}$$

*Proof.* At the true parameter, the partial outcome is

$$\boldsymbol{y}^\circ_{\beta_0} = \boldsymbol{y} - \beta_0\boldsymbol{x} = \boldsymbol{\mu}_0 + \boldsymbol{u} = \boldsymbol{Q}_0\boldsymbol{c}_0 + \boldsymbol{r}_0 + \boldsymbol{u}.$$

Apply the true-partition residual-maker $\boldsymbol{M}_0$. Since $\boldsymbol{Q}_0\boldsymbol{c}_0$ is in $\mathcal{N}_{K,\Lambda_0,T}$, Lemma 2 gives $\boldsymbol{M}_0\boldsymbol{Q}_0\boldsymbol{c}_0 = 0$. Therefore

$$\boldsymbol{M}_0\boldsymbol{y}^\circ_{\beta_0} = \boldsymbol{M}_0\boldsymbol{u} + \boldsymbol{M}_0\boldsymbol{r}_0.$$

The projected score is

$$\begin{aligned} S_T(\beta_0, \Lambda_0) &= \sqrt{T}\langle \boldsymbol{M}_0\boldsymbol{y}^\circ_{\beta_0}, \boldsymbol{M}_0\boldsymbol{w}\rangle_T \\ &= T^{-1/2}(\boldsymbol{M}_0\boldsymbol{u} + \boldsymbol{M}_0\boldsymbol{r}_0)'\boldsymbol{v}_0. \end{aligned}$$

Because $\boldsymbol{M}_0$ is symmetric and idempotent and $\boldsymbol{v}_0 = \boldsymbol{M}_0\boldsymbol{w}$, we have $\boldsymbol{M}_0\boldsymbol{v}_0 = \boldsymbol{v}_0$. Hence

$$(\boldsymbol{M}_0\boldsymbol{u})'\boldsymbol{v}_0 = \boldsymbol{u}'\boldsymbol{v}_0, \qquad (\boldsymbol{M}_0\boldsymbol{r}_0)'\boldsymbol{v}_0 = \boldsymbol{r}_0'\boldsymbol{v}_0.$$

Substituting these two identities gives (5.1). The term $T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0$ is handled by the score CLT and variance consistency. The term $T^{-1/2}\boldsymbol{r}_0'\boldsymbol{v}_0$ is the only approximation-rate term in this decomposition, because it is the only term containing the sieve residual $\boldsymbol{r}_0$. □

# 6 Omitted Breaks: False Predictivity as a Projection Drift

Let $\mathcal{N}_{K,0,T}$ be the stable/no-break nuisance space and let $\boldsymbol{M}_{\text{st}}$ be the residual-maker for orthogonal projection onto $\mathcal{N}_{K,0,T}$. Define

$$\boldsymbol{v}_{\text{st}} = \boldsymbol{M}_{\text{st}}\boldsymbol{w}, \qquad S_T^{\text{st}}(\beta) = \sqrt{T}\langle \boldsymbol{M}_{\text{st}}(\boldsymbol{y} - \beta\boldsymbol{x}), \boldsymbol{M}_{\text{st}}\boldsymbol{w}\rangle_T.$$

**Proposition 1** (Omitted-break projection drift)**.** *At the true parameter $\beta_0$,*

$$S_T^{\text{st}}(\beta_0) = T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_{\text{st}} + B_T, \qquad B_T = T^{-1/2}\boldsymbol{\mu}_0'\boldsymbol{v}_{\text{st}}. \tag{6.1}$$

*Equivalently,*

$$B_T = \sqrt{T}\langle \boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0, \boldsymbol{M}_{\text{st}}\boldsymbol{w}\rangle_T. \tag{6.2}$$

*If*

$$\langle \boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0, \boldsymbol{M}_{\text{st}}\boldsymbol{w}\rangle_T \to b \neq 0,$$

*then $B_T/\sqrt{T} \to b$, so the stable score has deterministic drift of order $\sqrt{T}$ at the true parameter.*

*Proof.* At $\beta_0$,

$$\boldsymbol{y} - \beta_0\boldsymbol{x} = \boldsymbol{\mu}_0 + \boldsymbol{u}.$$

Applying the stable residual-maker gives

$$\boldsymbol{M}_{\text{st}}(\boldsymbol{y} - \beta_0\boldsymbol{x}) = \boldsymbol{M}_{\text{st}}\boldsymbol{u} + \boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0.$$

Therefore

$$\begin{aligned} S_T^{\text{st}}(\beta_0) &= T^{-1/2}(\boldsymbol{M}_{\text{st}}\boldsymbol{u} + \boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0)'\boldsymbol{v}_{\text{st}} \\ &= T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_{\text{st}} + T^{-1/2}\boldsymbol{\mu}_0'\boldsymbol{v}_{\text{st}}, \end{aligned}$$

because $\boldsymbol{M}_{\text{st}}$ is symmetric and idempotent and $\boldsymbol{v}_{\text{st}} = \boldsymbol{M}_{\text{st}}\boldsymbol{w}$. This proves (6.1). The same symmetry and idempotence give

$$T^{-1/2}\boldsymbol{\mu}_0'\boldsymbol{v}_{\text{st}} = T^{-1/2}\boldsymbol{\mu}_0'\boldsymbol{M}_{\text{st}}\boldsymbol{w} = \sqrt{T}\langle \boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0, \boldsymbol{M}_{\text{st}}\boldsymbol{w}\rangle_T,$$

which proves (6.2). If the empirical inner product converges to $b \neq 0$, the drift is $B_T = \sqrt{T}\,b + o(\sqrt{T})$. □

As a toy example, suppose there are no covariates and the stable nuisance space contains only constants:

$$\mathcal{N}_{K,0,T} = \text{span}\{\mathbf{1}_T\}.$$

Let the true nuisance contain one level shift,

$$\mu_{0,t} = \delta\mathbf{1}\{t > k_0\}, \qquad \pi_T = T^{-1}\sum_{t=1}^{T}\mathbf{1}\{t > k_0\}.$$

The stable projection removes only the global mean. Hence

$$(\boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0)_t = \delta\{\mathbf{1}(t > k_0) - \pi_T\}.$$

If the residualized score direction is aligned with this step, for instance $\boldsymbol{M}_{\text{st}}\boldsymbol{w} = \boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0$, then

$$\langle \boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0, \boldsymbol{M}_{\text{st}}\boldsymbol{w}\rangle_T = \|\boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0\|_T^2 = \delta^2\pi_T(1-\pi_T).$$

When $\delta \neq 0$ and $\pi_T \to \pi \in (0,1)$, the limit is $\delta^2\pi(1-\pi) \neq 0$. The stable score therefore reports a false signal even at $\beta_0$.

The correct break-aware block sieve changes the projection target. Under the true partition,

$$\boldsymbol{\mu}_0 = \boldsymbol{Q}_0\boldsymbol{c}_0 + \boldsymbol{r}_0, \qquad \boldsymbol{Q}_0\boldsymbol{c}_0 \in \mathcal{N}_{K,\Lambda_0,T}, \qquad \|\boldsymbol{r}_0\|_T = o_p(1),$$

so $\boldsymbol{M}_0\boldsymbol{\mu}_0 = \boldsymbol{r}_0$ is small. Under the stable space, $\boldsymbol{M}_{\mathrm{st}}\boldsymbol{\mu}_0$ need not be small. The block-sieve projection removes the low-frequency break component before testing $\beta$; otherwise omitted nuisance breaks can enter the score as false predictivity.

# 7 Estimated Partition

The estimated break procedure chooses a random subspace

$$\widehat{\mathcal{N}}_T = \mathcal{N}_{K,\widehat{\Lambda},T}, \qquad \widehat{M}_T = \boldsymbol{M}_{K,\widehat{\Lambda}}.$$

The feasible score is

$$\widehat{S}_T(\beta_0) = \sqrt{T}\langle \widehat{M}_T(\boldsymbol{y} - \beta_0\boldsymbol{x}), \widehat{M}_T\boldsymbol{w}\rangle_T.$$

The true-partition score is

$$S_T^0(\beta_0) = \sqrt{T}\langle \boldsymbol{M}_0(\boldsymbol{y} - \beta_0\boldsymbol{x}), \boldsymbol{M}_0\boldsymbol{w}\rangle_T, \qquad \boldsymbol{M}_0 = \boldsymbol{M}_{K,\Lambda_0}.$$

Because each residual-maker is symmetric and idempotent,

$$\widehat{S}_T(\beta_0) = T^{-1/2}(\boldsymbol{\mu}_0 + \boldsymbol{u})'\widehat{M}_T\boldsymbol{w}, \qquad S_T^0(\beta_0) = T^{-1/2}(\boldsymbol{\mu}_0 + \boldsymbol{u})'\boldsymbol{M}_0\boldsymbol{w}.$$

Therefore the exact estimated-partition score error is

$$\begin{aligned} \widehat{S}_T(\beta_0) - S_T^0(\beta_0) = T^{-1/2}&\boldsymbol{\mu}_0'(\widehat{M}_T - \boldsymbol{M}_0)\boldsymbol{w} \\ &+ T^{-1/2}\boldsymbol{u}'(\widehat{M}_T - \boldsymbol{M}_0)\boldsymbol{w}. \end{aligned} \tag{7.1}$$

The first term is local sieve-bias perturbation. The second term is local noise perturbation. Thus the estimated-partition Wilks argument needs

$$T^{-1/2}\boldsymbol{\mu}_0'(\widehat{M}_T - \boldsymbol{M}_0)\boldsymbol{w} = o_p(1), \qquad T^{-1/2}\boldsymbol{u}'(\widehat{M}_T - \boldsymbol{M}_0)\boldsymbol{w} = o_p(1). \tag{7.2}$$

For the variance, the corresponding requirement is

$$\widehat{V}_T - V_T^0 = o_p(1), \qquad \widehat{V}_T = P_T g_t(\beta_0, \widehat{\Lambda})^2, \qquad V_T^0 = P_T g_t(\beta_0, \Lambda_0)^2. \tag{7.3}$$

Equations (7.1)–(7.3) are the exact bridge from estimated breaks to the true-partition geometry.

The reason this is weaker than operator-norm convergence is visible from (7.1). We do not need $\|\widehat{M}_T - \boldsymbol{M}_0\|_{op} \to 0$. We only need the difference to be small when applied to the score direction $\boldsymbol{w}$ and paired with $\boldsymbol{\mu}_0$ and $\boldsymbol{u}$, plus the analogous second-moment replacement.

The localization proof uses the RSS distance identity from Lemma 4. At $\beta_0$, set

$$\Delta_\Lambda = \boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0, \qquad \boldsymbol{y}^\circ_{\beta_0} = \boldsymbol{\mu}_0 + \boldsymbol{u}.$$

Then

$$\begin{aligned} RSS_K(\beta_0, \Lambda) - RSS_K(\beta_0, \Lambda_0) &= (\boldsymbol{y}^{\circ}_{\beta_0})' \Delta_\Lambda \boldsymbol{y}^{\circ}_{\beta_0} \\ &= \boldsymbol{\mu}_0' \Delta_\Lambda \boldsymbol{\mu}_0 + 2\boldsymbol{\mu}_0' \Delta_\Lambda \boldsymbol{u} + \boldsymbol{u}' \Delta_\Lambda \boldsymbol{u}. \end{aligned} \tag{7.4}$$

The deterministic term is the geometric drift from choosing the wrong nuisance subspace. The two stochastic terms must be uniformly smaller than that drift on wrong partitions. Part II states the primitive concentration conditions and then uses (7.4) to obtain break localization and order selection.

# 8 EL as a Finite-Sample Convex KKT Problem

For fixed $(K, \Lambda, \beta)$, everything in this section is finite-sample algebra in $L^2(P_T)$. Define the profiled residual, score direction, and scalar moment array

$$\boldsymbol{r}_\beta = \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y} - \beta \boldsymbol{x}), \qquad \boldsymbol{a} = \boldsymbol{M}_{K,\Lambda} \boldsymbol{w}, \qquad g_t(\beta, \Lambda) = r_{\beta,t} a_t.$$

Then

$$\Psi_T(\beta, \Lambda) = P_T g(\beta, \Lambda) = T^{-1} \sum_{t=1}^{T} g_t(\beta, \Lambda) = \langle \boldsymbol{r}_\beta, \boldsymbol{a} \rangle_T.$$

Thus the sieve basis $Q_{K,\Lambda}$ has already represented the nuisance feasible directions: $\mathrm{col}(Q_{K,\Lambda})$ is the tangent plane for nuisance variation, and $M_{K,\Lambda}$ keeps only the orthogonal score component.

EL adds one more finite-dimensional convex problem. The primal feasible set is the probability simplex cut by the affine moment hyperplane

$$\mathcal{C}_T(\beta, \Lambda) = \left\{ \pi \in \mathbb{R}^T_+ : \sum_{t=1}^{T} \pi_t = 1, \quad \sum_{t=1}^{T} \pi_t g_t(\beta, \Lambda) = 0 \right\}.$$

The empirical likelihood problem is equivalently

$$\min_{\pi \in \mathcal{C}_T(\beta, \Lambda)} - \sum_{t=1}^{T} \log(T \pi_t). \tag{8.1}$$

This is a convex barrier objective over a convex feasible set. Its Lagrangian is

$$\mathcal{L}(\pi, \eta, \lambda) = - \sum_{t=1}^{T} \log(T\pi_t) + \eta \left( \sum_{t=1}^{T} \pi_t - 1 \right) + T\lambda \sum_{t=1}^{T} \pi_t g_t.$$

The KKT first-order condition gives

$$-\frac{1}{\pi_t} + \eta + T\lambda g_t = 0.$$

After normalizing by $\sum_t \pi_t = 1$, the EL weights are

$$\widehat{\pi}_t(\lambda) = \frac{1}{T\{1 + \lambda g_t(\beta, \Lambda)\}}, \qquad 1 + \lambda g_t(\beta, \Lambda) > 0. \tag{8.2}$$

The remaining KKT equation is the dual moment equation

$$P_T\left[\frac{g_t(\beta,\Lambda)}{1+\lambda g_t(\beta,\Lambda)}\right]=0. \tag{8.3}$$

The log empirical likelihood ratio is therefore

$$\ell_T(\beta,\Lambda)=2\sum_{t=1}^{T}\log\{1+\lambda_T g_t(\beta,\Lambda)\}. \tag{8.4}$$

The quadratic EL reduction is also mostly KKT algebra. If $\max_t|\lambda_T g_t|=o_p(1)$, then expanding (8.3) gives

$$0=P_T\left[\frac{g_t}{1+\lambda_T g_t}\right]=P_T g_t-\lambda_T P_T g_t^2+R_{1T},\qquad R_{1T}=o_p(T^{-1/2}).$$

Hence, with $V_T=P_T g_t^2$,

$$\lambda_T=\frac{P_T g_t}{V_T}+o_p(T^{-1/2}).$$

Expanding (8.4) then yields

$$\ell_T(\beta_0,\Lambda)=\frac{T\{P_T g_t(\beta_0,\Lambda)\}^2}{V_T(\Lambda)}+o_p(1)=\frac{S_T(\beta_0,\Lambda)^2}{V_T(\Lambda)}+o_p(1). \tag{8.5}$$

The stochastic work is not to rediscover EL; it is only to verify the small size conditions under which the KKT expansion is valid.

# 9 Population Hilbert Space

## 9.1 Population object in $L^2(P)$

To express breaks in population geometry, include rescaled time $S\in(0,1]$. Let

$$O=(Y,X,W,S,U)$$

be a population random element with law $P$. The true regime map is

$$j_0(S)=j \quad\iff\quad \tau_{j,0}<S\le\tau_{j+1,0}.$$

The population model is the identity in $L^2(P)$:

$$Y=\beta_0X+\mu_0(S,W)+U,\qquad \mu_0(S,W)=\sum_{j=0}^{q_0}m_j(W)\mathbf{1}\{\tau_{j,0}<S\le\tau_{j+1,0}\}. \tag{9.1}$$

## 9.2 Population nuisance space

For a population partition $\Lambda$, define

$$\mathcal{N}^P_\Lambda = \overline{\left\{ \sum_{j=0}^{q} \mathbf{1}\{S \in I_j(\Lambda)\} h_j(W) : h_j \in L^2(P_W) \right\}}^{L^2(P)}.$$

Let $\Pi^P_\Lambda$ be the $L^2(P)$-orthogonal projection onto $\mathcal{N}^P_\Lambda$, and let

$$M^P_\Lambda = I - \Pi^P_\Lambda.$$

For $\beta \in \mathcal{B}$, define the population partial outcome

$$Y^\circ_\beta = Y - \beta X.$$

The population profiled residual and score direction are

$$r^P_\beta(\Lambda) = M^P_\Lambda Y^\circ_\beta = M^P_\Lambda (Y - \beta X), \qquad v^P(\Lambda) = M^P_\Lambda w.$$

The population moment is

$$\Psi(\beta, \Lambda) = \langle r^P_\beta(\Lambda), v^P(\Lambda) \rangle_P. \tag{9.2}$$

# 10 Population Identification

At the true population partition $\Lambda_0$, abbreviate

$$\mathcal{N}^P_0 = \mathcal{N}^P_{\Lambda_0}, \qquad \Pi^P_0 = \Pi^P_{\Lambda_0}, \qquad M^P_0 = M^P_{\Lambda_0}, \qquad v^P_0 = v^P(\Lambda_0) = M^P_0 w.$$

Because the true nuisance is regime-specific with the true population partition,

$$\mu_0(S, W) \in \mathcal{N}^P_0.$$

Therefore the true-partition population residual-maker removes it:

$$M^P_0 \mu_0 = 0. \tag{10.1}$$

This is the $L^2(P)$ analog of the finite-sample statement that the correct break-aware nuisance space absorbs the nuisance component.

Using the population model (9.1), the population partial outcome at a generic $\beta$ is

$$\begin{aligned} Y^\circ_\beta &= Y - \beta X \\ &= \mu_0(S, W) + U - (\beta - \beta_0) X. \end{aligned}$$

Projecting onto $(\mathcal{N}^P_0)^{\perp_P}$ gives

$$\begin{aligned} r^P_\beta(\Lambda_0) &= M^P_0 Y^\circ_\beta \\ &= M^P_0 \mu_0 + M^P_0 U - (\beta - \beta_0) M^P_0 X \\ &= M^P_0 U - (\beta - \beta_0) M^P_0 X, \end{aligned} \tag{10.2}$$

because $M_0^P \mu_0 = 0$. Taking the inner product with the true population score direction $v_0^P$ gives

$$\Psi(\beta, \Lambda_0) = \langle M_0^P U, v_0^P \rangle_P - (\beta - \beta_0)\langle M_0^P X, v_0^P \rangle_P. \tag{10.3}$$

This derivation is algebraic. The two population assumptions enter only after (10.3) is obtained.

The exogeneity condition is

$$\langle M_0^P U, v_0^P \rangle_P = 0.$$

The relevance condition is

$$\Gamma = \langle M_0^P X, v_0^P \rangle_P \neq 0.$$

Under these two conditions,

$$\Psi(\beta, \Lambda_0) = -(\beta - \beta_0)\Gamma.$$

Thus

$$\Psi(\beta, \Lambda_0) = 0 \quad \Longleftrightarrow \quad \beta = \beta_0. \tag{10.4}$$

# Part II
# Small Stochastic Bounds

## 11 Small Stochastic Bounds

Part I has already fixed the objects. For every candidate partition $\Lambda$, write

$$\boldsymbol{r}_\beta(\Lambda) = \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y} - \beta \boldsymbol{x}), \qquad \boldsymbol{a}(\Lambda) = \boldsymbol{M}_{K,\Lambda}\boldsymbol{w},$$

and

$$g_t(\beta, \Lambda) = r_{\beta,t}(\Lambda) a_t(\Lambda), \qquad \Psi_T(\beta, \Lambda) = P_T g(\beta, \Lambda) = \langle \boldsymbol{r}_\beta(\Lambda), \boldsymbol{a}(\Lambda)\rangle_T.$$

At the true partition, abbreviate

$$\boldsymbol{M}_0 = \boldsymbol{M}_{K,\Lambda_0}, \qquad \boldsymbol{v}_0 = \boldsymbol{M}_0 \boldsymbol{w}, \qquad \boldsymbol{y}_\beta^\circ = \boldsymbol{y} - \beta \boldsymbol{x}.$$

For the estimated partition, write

$$\widehat{\boldsymbol{M}} = \boldsymbol{M}_{K,\widehat{\Lambda}}, \qquad \widehat{\boldsymbol{v}} = \widehat{\boldsymbol{M}}\boldsymbol{w}.$$

The variance objects are

$$V_T(\Lambda) = P_T g_t(\beta_0, \Lambda)^2, \qquad V_T^0 = V_T(\Lambda_0), \qquad \widehat{V}_T = V_T(\widehat{\Lambda}).$$

The stochastic part proves the following targets one by one:

$$\sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \widehat{\Lambda}) - \Psi(\beta, \Lambda_0)| = o_p(1), \tag{11.1}$$

$$\sqrt{T}\Psi_T(\beta_0, \widehat{\Lambda}) \Rightarrow N(0, V), \qquad \widehat{V}_T \to_p V > 0,$$

and the local empirical-likelihood conditions that make the KKT expansion (8.5) valid. None of these statements changes the geometric objects. They only control their sampling error.

## 11.1 Standing stochastic conditions

Let $\mathcal{L}_T$ denote the admissible set of candidate partitions searched by the estimator. For local break-date arguments, let $\mathcal{N}_T(r_T) \subset \mathcal{L}_T$ denote the event/set of partitions whose break dates are within the localization radius $r_T$ of $\Lambda_0$. The following assumptions are the stochastic input for Part II. They are stated in the workbook notation so each proof below can be read directly.

**Assumption 1** (Empirical-to-population inner-product bridge)**.** *Uniformly over $\Lambda \in \mathcal{L}_T$, the empirical Gram matrices of the block sieve design are stable:*

$$\sup_{\Lambda \in \mathcal{L}_T} \left\| T^{-1} \boldsymbol{Q}'_{K,\Lambda} \boldsymbol{Q}_{K,\Lambda} - G_Q(\Lambda) \right\|_{op} = o_p(1),$$

*where*

$$0 < c \leq \lambda_{\min}\{G_Q(\Lambda)\} \leq \lambda_{\max}\{G_Q(\Lambda)\} \leq C < \infty \qquad \textit{uniformly over } \Lambda \in \mathcal{L}_T.$$

*The same uniform empirical laws hold for the projected cross moments involving $\boldsymbol{Q}_{K,\Lambda}$, $\boldsymbol{x}$, $\boldsymbol{w}$, and $\boldsymbol{u}$. Equivalently, for the finite sieve and break classes used here,*

$$\sup_{f,g \in \mathcal{F}_{K,T}} |\langle \boldsymbol{f}_T, \boldsymbol{g}_T \rangle_T - \langle f, g \rangle_P| = o_p(1).$$

*This is the bridge between $L^2(P_T)$ and $L^2(P)$. The spaces are not being identified as the same space; only the relevant inner products converge.*

**Assumption 2** (Score-negligible sieve remainder)**.** *For the true nuisance vector $\boldsymbol{\mu}_0$, the true-partition finite-sieve remainder*

$$\boldsymbol{r}_{\mu,T} = \boldsymbol{M}_0 \boldsymbol{\mu}_0$$

*satisfies*

$$\|\boldsymbol{r}_{\mu,T}\|_T = o_p(1), \qquad \sqrt{T}\,|\langle \boldsymbol{r}_{\mu,T}, \boldsymbol{v}_0 \rangle_T| = o_p(1), \qquad P_T\{r^2_{\mu,T,t} v^2_{0,t}\} = o_p(1).$$

*The first bound is the consistency-scale approximation bound. The second is the score-scale bias bound. The third removes the sieve remainder from the variance.*

**Assumption 3** (Estimated-partition primitive closure)**.** *The RSS break estimator localizes at radius $r_T$:*

$$\Pr\{\widehat{\Lambda} \in \mathcal{N}_T(r_T)\} \to 1.$$

*On $\mathcal{N}_T(r_T)$, local perturbations of the projected score and second moment are negligible:*

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} \sqrt{T}\,|\Psi_T(\beta_0, \Lambda) - \Psi_T(\beta_0, \Lambda_0)| = o_p(1),$$

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} |V_T(\Lambda) - V_T^0| = o_p(1),$$

*and, at the unscaled consistency level,*

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} \sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \Lambda) - \Psi_T(\beta, \Lambda_0)| = o_p(1).$$

*A sufficient local perturbation route is*

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} T^{-1/2} |\boldsymbol{r}'_{\mu,T}(\boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0)\boldsymbol{w}| = o_p(1),$$

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} T^{-1/2} |\boldsymbol{u}'(\boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0)\boldsymbol{w}| = o_p(1),$$

*with the variance replacement below. For a one-break local window, a standard sufficient rate is*

$$\zeta_K \sqrt{\frac{K r_T}{T}} \to 0,$$

*where $\zeta_K$ controls the sieve-envelope size.*

**Assumption 4** (RSS concentration and penalty rates)**.** *Let $\Delta_\Lambda = \boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0$. Outside the local set $\mathcal{N}_T(r_T)$, the deterministic RSS drift dominates stochastic RSS fluctuation:*

$$\inf_{\Lambda \in \mathcal{L}_T \setminus \mathcal{N}_T(r_T)} T^{-1} \boldsymbol{\mu}_0' \Delta_\Lambda \boldsymbol{\mu}_0 \geq c_T,$$

$$\sup_{\Lambda \in \mathcal{L}_T \setminus \mathcal{N}_T(r_T)} T^{-1} |2\boldsymbol{\mu}_0' \Delta_\Lambda \boldsymbol{u} + \boldsymbol{u}' \Delta_\Lambda \boldsymbol{u}| = o_p(c_T).$$

*One primitive route for the stochastic bound is conditionally sub-Gaussian innovations plus sieve entropy/concentration over the searched RSS class.*

*For unknown order, the underfitted deterministic loss and overfitted stochastic gain satisfy*

$$\inf_{q < q_0} \{RSS_K(q) - RSS_K(q_0)\} \geq cT\Delta_T^2 + o_p(T\Delta_T^2),$$

$$\sup_{q > q_0} \{RSS_K(q_0) - RSS_K(q)\} = O_p(K \log T),$$

*and the penalty obeys*

$$K \log T = o\{\mathrm{pen}_T(q, K)\}, \qquad \mathrm{pen}_T(q, K) = o(T\Delta_T^2)$$

*for every fixed nonzero order difference.*

**Assumption 5** (Projected LLN and CLT at the true partition)**.** *The true-partition projected empirical moments obey*

$$\langle \boldsymbol{M}_0 \boldsymbol{u}, \boldsymbol{v}_0 \rangle_T - \langle M_0^P U, v_0^P \rangle_P = o_p(1),$$

$$\langle \boldsymbol{M}_0 \boldsymbol{x}, \boldsymbol{v}_0 \rangle_T - \langle M_0^P X, v_0^P \rangle_P = o_p(1),$$

*and*

$$T^{-1/2} \boldsymbol{u}' \boldsymbol{v}_0 \Rightarrow N(0, V), \qquad 0 < V < \infty.$$

**Assumption 6** (Second and higher moment control). *With* $\tilde{g}_t^0 = (\boldsymbol{M}_0\boldsymbol{u})_t v_{0,t}$,

$$P_T(\tilde{g}_t^0)^2 \to_p V, \qquad P_T|g_t(\beta_0, \Lambda_0)|^3 = O_p(1), \qquad P_T|g_t(\beta_0, \widehat{\Lambda})|^3 = O_p(1),$$

*and*

$$\max_{t \le T} |g_t(\beta_0, \Lambda_0)|/\sqrt{T} = o_p(1), \qquad \max_{t \le T} |g_t(\beta_0, \widehat{\Lambda})|/\sqrt{T} = o_p(1).$$

**Assumption 7** (Scalar convex-hull condition). *With probability tending to one, both scalar samples* $\{g_t(\beta_0, \Lambda_0) : 1 \le t \le T\}$ *and* $\{g_t(\beta_0, \widehat{\Lambda}) : 1 \le t \le T\}$ *contain at least one strictly positive value and at least one strictly negative value.*

## 11.2 Primitive stochastic assumptions sufficient for Part II

The high-level assumptions above are implied by familiar primitive bounds. Let

$$v_{t,0}^* = v_{0,t} = (\boldsymbol{M}_0\boldsymbol{w})_t, \qquad \widehat{v}_{t,0}^* = (\widehat{\boldsymbol{M}}\boldsymbol{w})_t.$$

A useful sufficient primitive package is:

1. **Envelope control.**

$$\max_{t \le T} |v_{t,0}^*| = O_p(1 + \zeta_K), \qquad \max_{t \le T} |\widehat{v}_{t,0}^*| = O_p(1 + \zeta_K).$$

2. **Variance replacement.** If $\sigma_t^2 = \mathbb{E}(u_t^2 \mid \mathcal{F}_{t-1})$, then

$$T^{-1} \sum_{t=1}^{T} \sigma_t^2 \{(\widehat{v}_{t,0}^*)^2 - (v_{t,0}^*)^2\} = o_p(1).$$

3. **Local sieve-bias stability.**

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} T^{-1/2} |\boldsymbol{r}'_{\mu,T}(\boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0)\boldsymbol{w}| = o_p(1).$$

4. **Local noise perturbation.**

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} T^{-1/2} |\boldsymbol{u}'(\boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0)\boldsymbol{w}| = o_p(1),$$

   for example under the rate $\zeta_K\sqrt{K r_T/T} \to 0$ in the one-break local perturbation calculation.

5. **RSS concentration.** The errors have enough tail control, for example conditional sub-Gaussian tails, so the searched RSS class satisfies the concentration bounds in Assumption 4.

6. **Penalty separation.** The unknown-order penalty dominates searched overfit gains but is smaller than the underfit deterministic loss:

$$K \log T = o\{\mathrm{pen}_T(q, K)\}, \qquad \mathrm{pen}_T(q, K) = o(T\Delta_T^2).$$

These primitive conditions are not new geometry. They are the stochastic closure that makes the geometric KKT construction usable with estimated breaks.

## 11.3 Part II primitive proof-claim closure

This subsection records the detailed primitive route behind the high-level Part II assumptions. It is not a second model. It is the proof claim that the RSS localization, projected score CLT, variance replacement, max-score bound, and EL KKT expansion can all be obtained from standard local projection and concentration bounds.

Write

$$d_K = (q_0 + 1)K, \qquad b_K = 1 + \zeta_K, \qquad \mathbf{\Pi}_0 = I - \boldsymbol{M}_0.$$

At the true partition let $\boldsymbol{Q}_0 = \boldsymbol{Q}_{K,\Lambda_0}$. Assume the true nuisance has the finite-sieve decomposition

$$\boldsymbol{\mu}_0 = \boldsymbol{Q}_0 c_0 + \boldsymbol{r}_{\mu,T}.$$

**Lemma 7** (Scalar EL KKT proof claim)**.** *Let $Z_{t,T}$ be any scalar triangular moment array, and define*

$$A_T = T^{-1/2} \sum_{t=1}^{T} Z_{t,T}, \qquad B_T = T^{-1} \sum_{t=1}^{T} Z_{t,T}^2, \qquad m_T = \max_{t \le T} |Z_{t,T}|.$$

*If $A_T = O_p(1)$, $B_T \ge c + o_p(1)$, $m_T = o_p(\sqrt{T})$, and the scalar convex hull contains zero in its interior with probability tending to one, then the EL dual root $\lambda_T$ satisfies*

$$\lambda_T = \frac{T^{-1} \sum_t Z_{t,T}}{T^{-1} \sum_t Z_{t,T}^2} + o_p(T^{-1/2}),$$

*and*

$$2 \sum_{t=1}^{T} \log(1 + \lambda_T Z_{t,T}) = \frac{A_T^2}{B_T} + o_p(1).$$

*Proof.* The KKT equation is

$$0 = \sum_t \frac{Z_{t,T}}{1 + \lambda_T Z_{t,T}}, \qquad 1 + \lambda_T Z_{t,T} > 0.$$

With $S_1 = \sum_t Z_{t,T}$ and $S_2 = \sum_t Z_{t,T}^2$, the exact identity

$$S_1 = \lambda_T \sum_t \frac{Z_{t,T}^2}{1 + \lambda_T Z_{t,T}}$$

implies $|\lambda_T| = O_p(T^{-1/2})$ because $S_1 = O_p(\sqrt{T})$, $S_2 \ge cT + o_p(T)$, and $|S_1| m_T = o_p(T)$. Hence $|\lambda_T| m_T = o_p(1)$. Expanding the KKT equation around zero gives

$$0 = S_1 - \lambda_T S_2 + o_p(\sqrt{T}), \qquad \lambda_T = S_1/S_2 + o_p(T^{-1/2}).$$

Taylor expansion of the log ratio gives

$$2 \sum_t \log(1 + \lambda_T Z_{t,T}) = 2\lambda_T S_1 - \lambda_T^2 S_2 + o_p(1) = S_1^2/S_2 + o_p(1).$$

Since $S_1^2/S_2 = A_T^2/B_T$, the claim follows. □

**Lemma 8** (True-partition feasible-to-oracle proof claim)**.** *At $(\beta_0, \Lambda_0)$,*

$$\sqrt{T}\Psi_T(\beta_0, \Lambda_0) = T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0 + T^{-1/2}\boldsymbol{r}'_{\mu,T}\boldsymbol{v}_0.$$

*If $T^{-1/2}\boldsymbol{r}'_{\mu,T}\boldsymbol{v}_0 = o_p(1)$, then the score numerator is asymptotically the oracle numerator $T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0$. Moreover, if*

$$\|(\boldsymbol{M}_0\boldsymbol{r}_{\mu,T} - \boldsymbol{\Pi}_0\boldsymbol{u}) \odot \boldsymbol{v}_0\|_T = o_p(1), \qquad \|(\boldsymbol{M}_0\boldsymbol{r}_{\mu,T} - \boldsymbol{\Pi}_0\boldsymbol{u}) \odot \boldsymbol{v}_0\|_\infty = o_p(\sqrt{T}),$$

*then*

$$V_T^0 = T^{-1}\sum_{t=1}^{T} u_t^2 v_{0,t}^2 + o_p(1), \qquad \max_t |g_t(\beta_0, \Lambda_0)| = \max_t |u_t v_{0,t}| + o_p(\sqrt{T}).$$

*Proof.* Since $\boldsymbol{M}_0\boldsymbol{Q}_0 = 0$,

$$\boldsymbol{M}_0(\boldsymbol{y} - \beta_0\boldsymbol{x}) = \boldsymbol{M}_0(\boldsymbol{\mu}_0 + \boldsymbol{u}) = \boldsymbol{r}_{\mu,T} + \boldsymbol{M}_0\boldsymbol{u}.$$

Because $\boldsymbol{v}_0 = \boldsymbol{M}_0\boldsymbol{w}$ and $\boldsymbol{M}_0$ is symmetric and idempotent,

$$(\boldsymbol{M}_0\boldsymbol{u})'\boldsymbol{v}_0 = \boldsymbol{u}'\boldsymbol{v}_0.$$

This proves the numerator identity.

For the variance and max bounds, write

$$\boldsymbol{M}_0\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{\Pi}_0\boldsymbol{u}$$

and therefore

$$g_t(\beta_0, \Lambda_0) = u_t v_{0,t} + d_{t,0}, \qquad \boldsymbol{d}_0 = (\boldsymbol{M}_0\boldsymbol{r}_{\mu,T} - \boldsymbol{\Pi}_0\boldsymbol{u}) \odot \boldsymbol{v}_0.$$

The $L^2(P_T)$ bound on $\boldsymbol{d}_0$ gives the second-moment replacement by Cauchy-Schwarz, using $T^{-1}\sum_t u_t^2 v_{0,t}^2 = O_p(1)$. The sup-norm bound gives the max-score replacement. □

The preceding display is the place where primitive projection rates enter. A standard sufficient package is

$$\|\boldsymbol{M}_0\boldsymbol{r}_{\mu,T}\|_T = O_p(a_K), \qquad \|\boldsymbol{\Pi}_0\boldsymbol{u}\|_T = O_p\Big(\sqrt{d_K/T}\Big),$$

$$\|\boldsymbol{M}_0\boldsymbol{r}_{\mu,T}\|_\infty = O_p(a_K b_K), \qquad \|\boldsymbol{\Pi}_0\boldsymbol{u}\|_\infty = O_p\Big(\zeta_K\sqrt{d_K/T}\Big),$$

together with $\|\boldsymbol{v}_0\|_\infty = O_p(b_K)$. These imply

$$\|\boldsymbol{d}_0\|_T = O_p\Big(a_K b_K + b_K\sqrt{d_K/T}\Big) = o_p(1),$$

and

$$\|\boldsymbol{d}_0\|_\infty = O_p\Big(a_K b_K^2 + \zeta_K b_K\sqrt{d_K/T}\Big) = o_p(\sqrt{T}).$$

**Lemma 9** (Oracle score proof claim)**.** *Suppose either $\boldsymbol{v}_0$ is admissible for the martingale array, or there is a predictable approximation $v_{t,0}^{\mathrm{pr}}$ satisfying*

$$T^{-1/2}\sum_t u_t(v_{0,t}-v_{t,0}^{\mathrm{pr}})=o_p(1),$$

$$T^{-1}\sum_t u_t^2\{v_{0,t}^2-(v_{t,0}^{\mathrm{pr}})^2\}=o_p(1),\qquad \max_t|v_{t,0}^{\mathrm{pr}}|=O_p(b_K).$$

*If $b_K^{2+\delta_0}T^{-\delta_0/2}\to 0$ for some $\delta_0>0$, then*

$$T^{-1/2}\sum_t u_t v_{0,t}\Rightarrow N(0,V),$$

$$T^{-1}\sum_t u_t^2 v_{0,t}^2\to_p V,\qquad \max_t|u_t v_{0,t}|=o_p(\sqrt{T}).$$

*Proof.* Apply the martingale triangular-array CLT to $\xi_{t,T}=T^{-1/2}u_t v_{t,0}^{\mathrm{pr}}$. The conditional variance is

$$T^{-1}\sum_t \sigma_t^2(v_{t,0}^{\mathrm{pr}})^2\to_p V,$$

and the conditional Lyapunov term is bounded by

$$CT^{-(1+\delta_0/2)}\sum_t|v_{t,0}^{\mathrm{pr}}|^{2+\delta_0}\le Cb_K^{2+\delta_0}T^{-\delta_0/2}=o_p(1).$$

The predictable-approximation assumption transfers the CLT back to $\boldsymbol{v}_0$. The variance LLN follows by applying the same martingale moment argument to $(u_t^2-\sigma_t^2)(v_{t,0}^{\mathrm{pr}})^2$, and the max bound follows from

$$\mathbb{P}\{\max_t|u_t v_{t,0}^{\mathrm{pr}}|>\varepsilon\sqrt{T}\}\le C_\varepsilon b_K^{2+\delta_0}T^{-\delta_0/2}+o(1).$$

□

**Lemma 10** (Local estimated-partition bridge proof claim)**.** *For one break, let $\Lambda_0=(k_0)$, $\widetilde{\Lambda}=(\widetilde{k})$, and $h=|\widetilde{k}-k_0|=O_p(r_T)$. Under local Gram stability,*

$$T^{-1/2}\boldsymbol{\mu}_0'(\boldsymbol{M}_{K,\widetilde{k}}-\boldsymbol{M}_0)\boldsymbol{w}=O_p\left(\frac{b_K r_T}{\sqrt{T}}\right)+o_p(1),$$

*and*

$$T^{-1/2}\boldsymbol{u}'(\boldsymbol{M}_{K,\widetilde{k}}-\boldsymbol{M}_0)\boldsymbol{w}=O_p\left(\zeta_K\sqrt{\frac{Kr_T}{T}}\right)+o_p(1).$$

*If*

$$\frac{b_K r_T}{\sqrt{T}}\to 0,\qquad \zeta_K\sqrt{\frac{Kr_T}{T}}\to 0,$$

*then*

$$\sqrt{T}\{\Psi_T(\beta_0,\widetilde{\Lambda})-\Psi_T(\beta_0,\Lambda_0)\}=o_p(1).$$

*If also $\zeta_K^2 Kr_T/T\to 0$, then*

$$V_T(\widetilde{\Lambda})-V_T(\Lambda_0)=o_p(1).$$

*Proof.* Take $\widetilde{k} = k_0 + h$; the other side is symmetric. The only observations whose regime label changes are in $H = \{k_0 + 1, \ldots, k_0 + h\}$. Relative to the candidate partition, the true nuisance can be written as

$$\boldsymbol{\mu}_0 = \boldsymbol{Q}_{K,\widetilde{k}} c^{(\widetilde{k})} + \boldsymbol{d}_H + \boldsymbol{r}_{\mu,T},$$

where $\boldsymbol{d}_H$ is supported on $H$ and $\|\boldsymbol{d}_H\|_\infty = O_p(1)$. Since $\boldsymbol{M}_{K,\widetilde{k}} \boldsymbol{Q}_{K,\widetilde{k}} = 0$ and $\boldsymbol{M}_0 \boldsymbol{Q}_0 = 0$,

$$\boldsymbol{\mu}_0'(\boldsymbol{M}_{K,\widetilde{k}} - \boldsymbol{M}_0)\boldsymbol{w} = \boldsymbol{d}_H' \boldsymbol{M}_{K,\widetilde{k}} \boldsymbol{w} + \boldsymbol{r}_{\mu,T}'(\boldsymbol{M}_{K,\widetilde{k}} - \boldsymbol{M}_0)\boldsymbol{w}.$$

The first term is bounded by $\|\boldsymbol{d}_H\|_1 \|\boldsymbol{M}_{K,\widetilde{k}} \boldsymbol{w}\|_\infty = O_p(hb_K)$, and the second term is the local sieve-bias stability term. This proves the nuisance perturbation bound.

For the stochastic perturbation, write the residualized weights on the three blocks $A = \{1, \ldots, k_0\}$, $H$, and $B = \{k_0 + h + 1, \ldots, T\}$. The least-squares coefficient update on a long segment has the form

$$\widehat{\gamma}_{A\cup H} - \widehat{\gamma}_A = (P_{A\cup H}' P_{A\cup H})^{-1} P_H'(w_H - P_H \widehat{\gamma}_A),$$

so local Gram stability gives

$$\|\widehat{\gamma}_{A\cup H} - \widehat{\gamma}_A\| = O_p(h\zeta_K b_K/T),$$

and similarly on the right side. After cancellation, the remaining stochastic terms are controlled by

$$\|P_A' u_A\| + \|P_B' u_B\| = O_p(\sqrt{TK}), \qquad \|P_H' u_H\| = O_p(\zeta_K \sqrt{h}).$$

Multiplying by $T^{-1/2}$ gives the displayed $O_p\{\zeta_K \sqrt{K r_T/T}\}$ bound. The denominator replacement follows by the same local boundary argument: $v_{\widetilde{k}} - v_{k_0}$ is order $O_p(b_K)$ only on the $h$-point boundary and order $O_p(h\zeta_K^2 b_K/T)$ away from it, so

$$|V_T(\widetilde{\Lambda}) - V_T(\Lambda_0)| = O_p\left(\frac{b_K^2 r_T}{T} + \frac{\zeta_K^2 K r_T}{T}\right) + o_p(1).$$

The stated rates make this $o_p(1)$. A fixed finite number of breaks is handled by summing the same boundary calculation over the finitely many local neighborhoods. □

**Proposition 2** (Localization and order-selection proof claim)**.** *In the one-break calculation, suppose*

$$\Delta RSS_K(k) = RSS_K(\beta_0, k) - RSS_K(\beta_0, k_0)$$

*satisfies, uniformly for $h = |k - k_0|$,*

$$E\{\Delta RSS_K(k) \mid D_T\} \geq ch\Delta_T^2 - o_p(h\Delta_T^2),$$

*and*

$$|\Delta RSS_K(k) - E\{\Delta RSS_K(k) \mid D_T\}| = O_p(\sqrt{hK \log T}) + O_p(K \log T).$$

*Then with $R_T = K \log T$ and*

$$r_T = R_T \max\{\Delta_T^{-2}, \Delta_T^{-4}\},$$

*the known-order break estimator satisfies*

$$|\widehat{k} - k_0| = O_p(r_T).$$

*If $\Delta_T$ is bounded away from zero, this reduces to $|\widehat{k} - k_0| = O_p(K \log T)$. With fixed $\bar{q}$, if the overfit penalty increment dominates $K \log T$ and the underfit penalty saving is $o(T\Delta_T^2)$, then $\mathbb{P}(\widehat{q} = q_0) \to 1$.*

*Proof.* The RSS identity from the geometric part gives

$$\Delta RSS_K(k) = \boldsymbol{\mu}_0' A_k \boldsymbol{\mu}_0 + 2\boldsymbol{\mu}_0' A_k \boldsymbol{u} + \boldsymbol{u}' A_k \boldsymbol{u}, \qquad A_k = \boldsymbol{M}_{K,k} - \boldsymbol{M}_{K,k_0}.$$

Conditional centering and equal-rank residual-makers give $E(\boldsymbol{u}' A_k \boldsymbol{u} \mid D_T) = \sigma^2 \operatorname{tr}(A_k) = 0$, so the conditional mean is the deterministic drift. The displayed stochastic fluctuation bound comes from a sub-Gaussian bound for the linear term and a Hanson-Wright bound for the rank-$O(K)$ quadratic term, followed by a union bound over searched break dates.

For $h \geq M r_T$,

$$\frac{R_T}{h\Delta_T^2} \leq M^{-1}, \qquad \frac{\sqrt{hR_T}}{h\Delta_T^2} \leq M^{-1/2}.$$

Choosing $M$ large makes the stochastic part smaller than the positive drift, so $\Delta RSS_K(k) > 0$ uniformly outside the $Mr_T$ window with probability tending to one. Since $\widehat{k}$ minimizes RSS and $k_0$ is admissible, $|\widehat{k} - k_0| = O_p(r_T)$.

For unknown order, underfitting loses at least $cT\Delta_T^2 + o_p(T\Delta_T^2)$ in deterministic RSS, while the penalty saving is $o(T\Delta_T^2)$. Overfitting can improve RSS only by $O_p(K \log T)$, while the penalty increase is larger than $K \log T$. A finite union over $q \leq \bar{q}$ gives $\mathbb{P}(\widehat{q} = q_0) \to 1$. □

## 11.4 RSS localization and order selection

**Proposition 3** (RSS localization)**.** *Under Assumption 4, the known-order RSS estimator satisfies*

$$\Pr\{\widehat{\Lambda} \in \mathcal{N}_T(r_T)\} \to 1.$$

*Proof.* For any candidate $\Lambda$, Section 7 gives the exact decomposition

$$RSS_K(\beta_0, \Lambda) - RSS_K(\beta_0, \Lambda_0) = \boldsymbol{\mu}_0' \Delta_\Lambda \boldsymbol{\mu}_0 + 2\boldsymbol{\mu}_0' \Delta_\Lambda \boldsymbol{u} + \boldsymbol{u}' \Delta_\Lambda \boldsymbol{u},$$

where $\Delta_\Lambda = \boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0$. On $\mathcal{L}_T \setminus \mathcal{N}_T(r_T)$, Assumption 4 gives

$$T^{-1} \boldsymbol{\mu}_0' \Delta_\Lambda \boldsymbol{\mu}_0 \geq c_T$$

uniformly, while

$$T^{-1} |2\boldsymbol{\mu}_0' \Delta_\Lambda \boldsymbol{u} + \boldsymbol{u}' \Delta_\Lambda \boldsymbol{u}| = o_p(c_T)$$

uniformly. Therefore

$$\inf_{\Lambda \in \mathcal{L}_T \setminus \mathcal{N}_T(r_T)} \{RSS_K(\beta_0, \Lambda) - RSS_K(\beta_0, \Lambda_0)\} > 0$$

with probability tending to one. A minimizer over the known-order search set cannot lie outside $\mathcal{N}_T(r_T)$ on this event. □

**Proposition 4** (Unknown-order penalty selection). *Under Assumption 4, the penalized RSS selector satisfies*

$$\Pr(\widehat{q} = q_0) \to 1.$$

*Proof.* For underfitting, Assumption 4 gives

$$\inf_{q<q_0} \{RSS_K(q) - RSS_K(q_0)\} \geq cT\Delta_T^2 + o_p(T\Delta_T^2).$$

The penalty difference for a fixed lower order is $o(T\Delta_T^2)$, so the deterministic loss from omitting a true break dominates any penalty saving. Thus underfitted orders are rejected with probability tending to one.

For overfitting,

$$\sup_{q>q_0}\{RSS_K(q_0) - RSS_K(q)\} = O_p(K \log T).$$

The penalty increment satisfies $K \log T = o\{\mathrm{pen}_T(q, K)\}$, so any spurious RSS gain is dominated by the penalty increase with probability tending to one. Thus overfitted orders are rejected with probability tending to one. Combining the underfit and overfit cases gives $\Pr(\widehat{q} = q_0) \to 1$. $\square$

## 11.5 Gram stability

**Lemma 11** (Gram stability gives well-behaved projections). *Under Assumption 1, uniformly over $\Lambda \in \mathcal{L}_T$,*

$$\lambda_{\min}\{T^{-1}\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda}\} \geq c/2$$

*with probability tending to one. Hence the coordinate formula for $\Pi_{K,\Lambda,T}$ is stable on $\mathcal{L}_T$, and the residual-maker $\boldsymbol{M}_{K,\Lambda} = I - \Pi_{K,\Lambda,T}$ remains an orthogonal contraction in $L^2(P_T)$:*

$$\|\boldsymbol{M}_{K,\Lambda}\boldsymbol{z}\|_T \leq \|\boldsymbol{z}\|_T \qquad \text{for every } \boldsymbol{z} \in L^2(P_T).$$

*Proof.* By Weyl's eigenvalue inequality,

$$\lambda_{\min}\{T^{-1}\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda}\} \geq \lambda_{\min}\{G_Q(\Lambda)\} - \left\|T^{-1}\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda} - G_Q(\Lambda)\right\|_{op}.$$

Taking the infimum over $\Lambda \in \mathcal{L}_T$, Assumption 1 puts the first term above $c$ and the second term below $c/2$ with probability tending to one. This proves the eigenvalue bound.

On that event, the least-squares projection has the usual stable coordinate form

$$\Pi_{K,\Lambda,T} = \boldsymbol{Q}_{K,\Lambda}(\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda})^{-1}\boldsymbol{Q}'_{K,\Lambda}.$$

Independently of coordinates, $\Pi_{K,\Lambda,T}$ is the orthogonal projection onto $\mathcal{N}_{K,\Lambda,T}$. Therefore $\boldsymbol{M}_{K,\Lambda} = I - \Pi_{K,\Lambda,T}$ is also an orthogonal projection. Orthogonal projections are contractions in Hilbert norm. $\square$

## 11.6 Empirical-to-population projection bridge

**Lemma 12** (Inner-product bridge from $L^2(P_T)$ to $L^2(P)$)**.** *Under Assumption 1, uniformly over the sieve and break classes used in the projected score,*

$$\langle \boldsymbol{f}_T, \boldsymbol{g}_T \rangle_T = \langle f, g \rangle_P + o_p(1).$$

*In particular,*

$$\sup_{\Lambda \in \mathcal{L}_T} \left\| T^{-1} \boldsymbol{Q}'_{K,\Lambda} \boldsymbol{Q}_{K,\Lambda} - G_Q(\Lambda) \right\|_{op} = o_p(1),$$

*and the corresponding projected $(x, w, u)$ cross moments converge to their population limits.*

*Proof.* The first display is the uniform empirical law in Assumption 1. Taking $f$ and $g$ to be two coordinates of the block sieve row gives the displayed Gram convergence. Taking one or both functions to be the projected $x$, $w$, or $u$ directions gives the corresponding cross-moment convergence. This is exactly the bridge needed by the workbook: finite-sample projection geometry is written in $L^2(P_T)$, while identification is written in $L^2(P)$, and the common object is convergence of the relevant inner products. □

## 11.7 Sieve remainder is small

**Lemma 13** (True-partition sieve remainder is negligible)**.** *Under Assumption 2,*

$$\langle \boldsymbol{M}_0 \boldsymbol{\mu}_0, \boldsymbol{v}_0 \rangle_T = o_p(1), \qquad \sqrt{T} \langle \boldsymbol{M}_0 \boldsymbol{\mu}_0, \boldsymbol{v}_0 \rangle_T = o_p(1),$$

*and*

$$V_T^0 = P_T \{ (\boldsymbol{M}_0 \boldsymbol{u})_t v_{0,t} \}^2 + o_p(1).$$

*Proof.* By definition $\boldsymbol{r}_{\mu,T} = \boldsymbol{M}_0 \boldsymbol{\mu}_0$. The first two displays are exactly the consistency-scale and score-scale components of Assumption 2.

At $\beta_0$,

$$\boldsymbol{M}_0 \boldsymbol{y}^{\circ}_{\beta_0} = \boldsymbol{M}_0 (\boldsymbol{\mu}_0 + \boldsymbol{u}) = \boldsymbol{r}_{\mu,T} + \boldsymbol{M}_0 \boldsymbol{u}.$$

Thus

$$g_t(\beta_0, \Lambda_0) = \{ (\boldsymbol{M}_0 \boldsymbol{u})_t + r_{\mu,T,t} \} v_{0,t}.$$

Let $\tilde{g}_t^0 = (\boldsymbol{M}_0 \boldsymbol{u})_t v_{0,t}$ and $d_t = r_{\mu,T,t} v_{0,t}$. Then

$$V_T^0 - P_T (\tilde{g}_t^0)^2 = 2 P_T \tilde{g}_t^0 d_t + P_T d_t^2.$$

The final term is $o_p(1)$ by Assumption 2. The cross term is bounded by Cauchy-Schwarz:

$$|P_T \tilde{g}_t^0 d_t| \le \{ P_T (\tilde{g}_t^0)^2 \}^{1/2} \{ P_T d_t^2 \}^{1/2} = o_p(1),$$

using the bounded second moment in Assumption 6. Hence $V_T^0 = P_T (\tilde{g}_t^0)^2 + o_p(1)$. □

## 11.8 Break replacement

**Lemma 14** (Estimated partition may replace the true partition)**.** *Under Assumption 3,*

$$\sup_{\beta\in\mathcal{B}} |\Psi_T(\beta,\widehat{\Lambda}) - \Psi_T(\beta,\Lambda_0)| = o_p(1),$$

$$\sqrt{T}\,|\Psi_T(\beta_0,\widehat{\Lambda}) - \Psi_T(\beta_0,\Lambda_0)| = o_p(1), \qquad \widehat{V}_T - V_T^0 = o_p(1).$$

*Proof.* By Assumption 3, $\Pr\{\widehat{\Lambda} \in \mathcal{N}_T(r_T)\} \to 1$. On this event, $\widehat{\Lambda}$ is one of the partitions over which the three local suprema in the same assumption are taken. Therefore

$$\sup_{\beta\in\mathcal{B}} |\Psi_T(\beta,\widehat{\Lambda}) - \Psi_T(\beta,\Lambda_0)| \le \sup_{\Lambda\in\mathcal{N}_T(r_T)} \sup_{\beta\in\mathcal{B}} |\Psi_T(\beta,\Lambda) - \Psi_T(\beta,\Lambda_0)| = o_p(1),$$

$$\sqrt{T}|\Psi_T(\beta_0,\widehat{\Lambda}) - \Psi_T(\beta_0,\Lambda_0)| \le \sup_{\Lambda\in\mathcal{N}_T(r_T)} \sqrt{T}|\Psi_T(\beta_0,\Lambda) - \Psi_T(\beta_0,\Lambda_0)| = o_p(1),$$

and

$$|\widehat{V}_T - V_T^0| \le \sup_{\Lambda\in\mathcal{N}_T(r_T)} |V_T(\Lambda) - V_T^0| = o_p(1).$$

The conclusion follows after adding the negligible probability of the complement of the localization event. ☐

## 11.9 Uniform moment bridge

**Lemma 15** (Uniform projected moment bridge)**.** *Under Assumptions 2, 3, and 5,*

$$\sup_{\beta\in\mathcal{B}} |\Psi_T(\beta,\widehat{\Lambda}) - \Psi(\beta,\Lambda_0)| = o_p(1). \tag{11.1}$$

*Proof.* First prove the bridge at the true partition. Since $\boldsymbol{y}_\beta^\circ = \boldsymbol{\mu}_0 + \boldsymbol{u} - (\beta-\beta_0)\boldsymbol{x}$,

$$\begin{aligned}\Psi_T(\beta,\Lambda_0) &= \langle \boldsymbol{M}_0\boldsymbol{y}_\beta^\circ, \boldsymbol{v}_0\rangle_T \\ &= \langle \boldsymbol{M}_0\boldsymbol{\mu}_0, \boldsymbol{v}_0\rangle_T + \langle \boldsymbol{M}_0\boldsymbol{u}, \boldsymbol{v}_0\rangle_T - (\beta-\beta_0)\langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{v}_0\rangle_T.\end{aligned}$$

The population moment from Part I is

$$\Psi(\beta,\Lambda_0) = \langle M_0^P U, v_0^P\rangle_P - (\beta-\beta_0)\langle M_0^P X, v_0^P\rangle_P.$$

Subtracting gives

$$\begin{aligned}\Psi_T(\beta,\Lambda_0) - \Psi(\beta,\Lambda_0) = {} & \langle \boldsymbol{M}_0\boldsymbol{\mu}_0, \boldsymbol{v}_0\rangle_T \\ & + \left\{\langle \boldsymbol{M}_0\boldsymbol{u}, \boldsymbol{v}_0\rangle_T - \langle M_0^P U, v_0^P\rangle_P\right\} \\ & - (\beta-\beta_0)\left\{\langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{v}_0\rangle_T - \langle M_0^P X, v_0^P\rangle_P\right\}.\end{aligned}$$

The first term is $o_p(1)$ by Lemma 13. The next two bracketed terms are $o_p(1)$ by Assumption 5. Because $\mathcal{B}$ is fixed and bounded, the convergence is uniform in $\beta \in \mathcal{B}$:

$$\sup_{\beta\in\mathcal{B}} |\Psi_T(\beta,\Lambda_0) - \Psi(\beta,\Lambda_0)| = o_p(1).$$

Finally, Lemma 14 gives

$$\sup_{\beta\in\mathcal{B}} |\Psi_T(\beta,\widehat{\Lambda}) - \Psi_T(\beta,\Lambda_0)| = o_p(1).$$

The triangle inequality proves the stated bridge. ☐

## 11.10 Score CLT

**Lemma 16** (Projected score CLT)**.** *Under Assumptions 2, 3, and 5,*

$$\sqrt{T}\Psi_T(\beta_0, \widehat{\Lambda}) \Rightarrow N(0, V).$$

*Proof.* At the true partition,

$$\begin{aligned}\sqrt{T}\Psi_T(\beta_0, \Lambda_0) &= \sqrt{T}\langle \boldsymbol{M}_0 \boldsymbol{y}^\circ_{\beta_0}, \boldsymbol{v}_0\rangle_T \\ &= T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0 + \sqrt{T}\langle \boldsymbol{M}_0 \boldsymbol{\mu}_0, \boldsymbol{v}_0\rangle_T.\end{aligned}$$

The first equality is the definition of $\Psi_T$. The second uses $\boldsymbol{y}^\circ_{\beta_0} = \boldsymbol{\mu}_0 + \boldsymbol{u}$ and the symmetry/idempotence of $\boldsymbol{M}_0$, so $\langle \boldsymbol{M}_0\boldsymbol{u}, \boldsymbol{v}_0\rangle_T = T^{-1}\boldsymbol{u}'\boldsymbol{v}_0$. The approximation-bias term is $o_p(1)$ by Assumption 2. The leading term $T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0$ converges to $N(0, V)$ by Assumption 5. Therefore

$$\sqrt{T}\Psi_T(\beta_0, \Lambda_0) \Rightarrow N(0, V).$$

Lemma 14 gives

$$\sqrt{T}\{\Psi_T(\beta_0, \widehat{\Lambda}) - \Psi_T(\beta_0, \Lambda_0)\} = o_p(1).$$

Slutsky's theorem gives the result for $\widehat{\Lambda}$. □

## 11.11 Variance consistency

**Lemma 17** (Projected variance consistency)**.** *Under Assumptions 2, 3, and 6,*

$$\widehat{V}_T \to_p V > 0.$$

*Proof.* Lemma 13 gives

$$V_T^0 = P_T(\tilde{g}_t^0)^2 + o_p(1), \qquad \tilde{g}_t^0 = (\boldsymbol{M}_0\boldsymbol{u})_t v_{0,t}.$$

Assumption 6 gives $P_T(\tilde{g}_t^0)^2 \to_p V$. Therefore $V_T^0 \to_p V$. Lemma 14 gives $\widehat{V}_T - V_T^0 = o_p(1)$, so $\widehat{V}_T \to_p V$. The lower bound $V > 0$ is part of Assumption 5; it rules out a degenerate residual score direction. □

## 11.12 Max-score and convex hull

**Lemma 18** (EL local feasibility and max-score)**.** *Under Assumptions 6 and 7,*

$$\max_{t \le T} |g_t(\beta_0, \Lambda_0)|/\sqrt{T} = o_p(1), \qquad \max_{t \le T} |g_t(\beta_0, \widehat{\Lambda})|/\sqrt{T} = o_p(1),$$

*and the origin is in the interior of the scalar convex hull for both* $\{g_t(\beta_0, \Lambda_0) : 1 \le t \le T\}$ *and* $\{g_t(\beta_0, \widehat{\Lambda}) : 1 \le t \le T\}$ *with probability tending to one.*

*Proof.* The two max-score bounds are displayed in Assumption 6. For scalar moments, the convex hull of the observed values is the interval from the sample minimum to the sample maximum. The origin is in its interior exactly when the minimum is strictly negative and the maximum is strictly positive. Assumption 7 states that this sign condition holds with probability tending to one for both the known-partition and estimated-partition moment arrays. □

## 11.13 EL KKT local expansion

**Lemma 19** (Local EL KKT expansion)**.** *Under Assumptions 2, 3, 5, 6, and 7, with* $g_t = g_t(\beta_0, \widehat{\Lambda})$,

$$\lambda_T = \frac{P_T g_t}{P_T g_t^2} + o_p(T^{-1/2}), \qquad \max_{t \le T} |\lambda_T g_t| = o_p(1),$$

*and*

$$\ell_T(\beta_0, \widehat{\Lambda}) = \frac{T\{\Psi_T(\beta_0, \widehat{\Lambda})\}^2}{\widehat{V}_T} + o_p(1).$$

*Proof.* By Lemma 18, the scalar convex-hull condition holds with probability tending to one, so the EL dual root $\lambda_T$ exists in the interior region where $1 + \lambda_T g_t > 0$. By Lemma 16,

$$P_T g_t = \Psi_T(\beta_0, \widehat{\Lambda}) = O_p(T^{-1/2}).$$

By Lemma 17,

$$P_T g_t^2 = \widehat{V}_T \to_p V > 0.$$

Expanding the KKT equation

$$0 = P_T \left[ \frac{g_t}{1 + \lambda_T g_t} \right]$$

around $\lambda_T = 0$ gives

$$0 = P_T g_t - \lambda_T P_T g_t^2 + O_p\{\lambda_T^2 P_T |g_t|^3\}.$$

The third-moment bound in Assumption 6 implies the remainder is $O_p(\lambda_T^2)$. Since $P_T g_t = O_p(T^{-1/2})$ and $P_T g_t^2 \to_p V > 0$, the root satisfies $\lambda_T = O_p(T^{-1/2})$ and hence

$$\lambda_T = \frac{P_T g_t}{P_T g_t^2} + o_p(T^{-1/2}).$$

Combining $\lambda_T = O_p(T^{-1/2})$ with the max-score bound gives

$$\max_{t \le T} |\lambda_T g_t| \le |\lambda_T| \max_{t \le T} |g_t| = o_p(1).$$

Therefore the log EL ratio can be Taylor-expanded:

$$\begin{aligned}
\ell_T(\beta_0, \widehat{\Lambda}) &= 2 \sum_{t=1}^{T} \log(1 + \lambda_T g_t) \\
&= 2T \lambda_T P_T g_t - T \lambda_T^2 P_T g_t^2 + O_p\{T |\lambda_T|^3 P_T |g_t|^3\}.
\end{aligned}$$

The final term is $o_p(1)$, because $\lambda_T = O_p(T^{-1/2})$ and $P_T |g_t|^3 = O_p(1)$. Substituting $\lambda_T = P_T g_t / P_T g_t^2 + o_p(T^{-1/2})$ yields

$$\ell_T(\beta_0, \widehat{\Lambda}) = \frac{T(P_T g_t)^2}{P_T g_t^2} + o_p(1) = \frac{T\{\Psi_T(\beta_0, \widehat{\Lambda})\}^2}{\widehat{V}_T} + o_p(1).$$

$\square$

### 11.14 Known-partition Wilks

**Proposition 5** (Known-partition profile EL Wilks)**.** *Under the true partition $\Lambda_0$, the profile empirical likelihood ratio satisfies*

$$\ell_T(\beta_0, \Lambda_0) \Rightarrow \chi_1^2.$$

*Proof.* The true-partition versions of Lemmas 16 and 17 give

$$\sqrt{T}\Psi_T(\beta_0, \Lambda_0) \Rightarrow N(0, V), \qquad V_T^0 \to_p V > 0.$$

The max-score and convex-hull parts of Assumptions 6 and 7 apply to $g_t(\beta_0, \Lambda_0)$, so the KKT expansion from Part I gives

$$\ell_T(\beta_0, \Lambda_0) = \frac{T\{\Psi_T(\beta_0, \Lambda_0)\}^2}{V_T^0} + o_p(1).$$

Therefore

$$\ell_T(\beta_0, \Lambda_0) = \left\{ \frac{\sqrt{T}\Psi_T(\beta_0, \Lambda_0)}{\sqrt{V_T^0}} \right\}^2 + o_p(1) \Rightarrow \chi_1^2.$$

□

### 11.15 Part II consequences

The uniform bridge gives the consistency input. If $\widehat{\beta}_T$ is an approximate sample root,

$$\Psi_T(\widehat{\beta}_T, \widehat{\Lambda}) = o_p(1),$$

then Lemma 15 gives

$$\Psi(\widehat{\beta}_T, \Lambda_0) = o_p(1).$$

Part I gives the unique population zero, so $\widehat{\beta}_T \to_p \beta_0$.

For Wilks, Lemmas 16, 17, and 19 give

$$\ell_T(\beta_0, \widehat{\Lambda}) = \frac{T\{\Psi_T(\beta_0, \widehat{\Lambda})\}^2}{\widehat{V}_T} + o_p(1), \qquad \frac{\sqrt{T}\Psi_T(\beta_0, \widehat{\Lambda})}{\sqrt{\widehat{V}_T}} \Rightarrow N(0, 1).$$

Part III only merges these two displayed statements with the geometric KKT reduction already proved in Part I.

# Part III
# Merging Part

## 12 Merging Part: Moment Bridge, EL KKT, and $\chi_1^2$

The merge step should not say that the unscaled moment bridge is itself equivalent to Wilks. The correct statement is more precise:

The same projected moment $\Psi_T$ is the interface for both targets. Unscaled convergence of $\Psi_T$ to $\Psi$ merges with population uniqueness to prove consistency. Scaled convergence of $\Psi_T$, together with the EL KKT quadratic reduction, merges to prove $\chi^2_1$.

## 12.1 Merge A: consistency

Part I gives the population geometry

$$\Psi(\beta, \Lambda_0) = \langle M_0^P Y_\beta^\circ, v_0^P \rangle_P = -(\beta - \beta_0)\Gamma$$

under exogeneity and relevance. Hence

$$\Psi(\beta, \Lambda_0) = 0 \quad \Longleftrightarrow \quad \beta = \beta_0.$$

Part II gives the stochastic bridge

$$\sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \widehat{\Lambda}) - \Psi(\beta, \Lambda_0)| = o_p(1).$$

Therefore any approximate sample root satisfies

$$\Psi_T(\widehat{\beta}_T, \widehat{\Lambda}) = o_p(1) \quad \Longrightarrow \quad \widehat{\beta}_T \to_p \beta_0.$$

This is the estimation target.

## 12.2 Merge B: Wilks inference

For inference, Part I gives the EL KKT representation

$$\ell_T(\beta, \widehat{\Lambda}) = 2 \sum_{t=1}^{T} \log\{1 + \lambda_T g_t(\beta, \widehat{\Lambda})\},$$

where $\lambda_T$ solves

$$P_T \left[ \frac{g_t(\beta, \widehat{\Lambda})}{1 + \lambda_T g_t(\beta, \widehat{\Lambda})} \right] = 0.$$

Part II verifies the local size and convex-hull conditions, so the KKT equation can be expanded at $\beta = \beta_0$:

$$\ell_T(\beta_0, \widehat{\Lambda}) = \frac{T\{\Psi_T(\beta_0, \widehat{\Lambda})\}^2}{V_T(\widehat{\Lambda})} + o_p(1).$$

Part II also gives

$$\frac{\sqrt{T}\Psi_T(\beta_0, \widehat{\Lambda})}{\sqrt{V_T(\widehat{\Lambda})}} \Rightarrow N(0, 1).$$

Combining the last two displays gives

$$\ell_T(\beta_0, \widehat{\Lambda}) \Rightarrow \chi^2_1. \tag{12.1}$$

Thus the "equivalence" is conditional on the geometric KKT reduction. Once EL has been reduced to the square of the standardized projected moment, Wilks is equivalent to the standardized score CLT. It is not equivalent to the unscaled LLN bridge used for consistency.

**Proof-body split.** Geometry owns the model, nuisance tangent space, projection KKT, residual score, population zero, and EL primal/dual KKT. Small stochastic bounds only verify convergence and local size. The merging part then turns the same projected moment into either consistency or Wilks.

# 13 Optional Semiparametric Efficiency Theorem With Efficient Weight

This optional block comes after the Wilks argument and is not part of the minimum profile-sieve Wilks proof. The generic projected score with a residualized instrument $w$ gives valid orthogonal EL inference under the stochastic conditions above. A Fisher-style semiparametric efficiency claim needs an extra local likelihood model and the efficient residualized instrument.

The logic is: choose $w = w^*$, prove efficient-score equivalence, then derive the asymptotic linear expansion. The homoskedastic case reduces to $w = X$; the heteroskedastic oracle case uses $w^* = X/\sigma^2(A)$.

Let

$$A = (J_0, W),$$

where $J_0$ is the true regime indicator. If the paper keeps the notation $A = (S, W)$, then every conditional projection $E(\cdot \mid A)$ below should be read as projection onto the true break-aware nuisance sigma-field generated by the true regime and $W$. This avoids treating the nuisance space as an arbitrary continuous-time $S$-indexed space. The population model is

$$Y = \beta_0 X + \mu_0(A) + U, \qquad \mu_0(A) \in \mathcal{N}_0^P = \mathcal{N}_{\Lambda_0}^P.$$

For the efficiency calculation, assume the conditional Gaussian location model

$$U \mid X, A \sim N(0, \sigma^2(A)), \qquad 0 < c \le \sigma^2(A) \le C < \infty.$$

The homoskedastic case is the special case $\sigma^2(A) = \sigma^2$. The population nuisance tangent directions generated by paths $\mu_t(A) = \mu_0(A) + th(A)$, $h \in \mathcal{N}_0^P$, are

$$\mathcal{T}_\mu = \overline{\left\{ \frac{Uh(A)}{\sigma^2(A)} : h \in \mathcal{N}_0^P \right\}}^{L^2(P)}.$$

The raw score for $\beta$ is

$$S_\beta^0 = \frac{UX}{\sigma^2(A)}.$$

**Proposition 6** (Efficient score)**.** *Let*

$$m_X(A) = E(X \mid A), \qquad \widetilde{X} = X - m_X(A).$$

*The efficient score for $\beta$ is*

$$S_{\text{eff}} = \frac{U\widetilde{X}}{\sigma^2(A)},$$

*and the efficient information is*

$$I_{\text{eff}} = E(S_{\text{eff}}^2) = E\left\{\frac{\widetilde{X}^2}{\sigma^2(A)}\right\}.$$

*Proof.* The projection of $S_\beta^0$ onto $\mathcal{T}_\mu$ has the form $Uh^*(A)/\sigma^2(A)$. The function $h^*$ minimizes

$$E\left[\left\{\frac{U}{\sigma^2(A)}\right\}^2 \{X - h(A)\}^2\right].$$

Since $E(U^2 \mid X, A) = \sigma^2(A)$, this criterion is

$$E\left[\frac{\{X - h(A)\}^2}{\sigma^2(A)}\right].$$

Because $\sigma^2(A)$ is $A$-measurable, conditional minimization gives $h^*(A) = E(X \mid A) = m_X(A)$. Therefore

$$\Pi(S_\beta^0 \mid \mathcal{T}_\mu) = \frac{Um_X(A)}{\sigma^2(A)},$$

and

$$S_{\text{eff}} = S_\beta^0 - \Pi(S_\beta^0 \mid \mathcal{T}_\mu) = \frac{U\{X - m_X(A)\}}{\sigma^2(A)}.$$

Finally,

$$E(S_{\text{eff}}^2) = E\left\{\frac{U^2\widetilde{X}^2}{\sigma^4(A)}\right\} = E\left\{\frac{\widetilde{X}^2}{\sigma^2(A)}\right\},$$

again using $E(U^2 \mid X, A) = \sigma^2(A)$. □

The efficient population weight is

$$w^*(O) = \frac{X}{\sigma^2(A)}.$$

Since $\sigma^2(A)$ is $A$-measurable,

$$E(w^* \mid A) = \frac{E(X \mid A)}{\sigma^2(A)}.$$

Thus the population residualized efficient instrument is

$$M_0^P w^* = w^* - E(w^* \mid A) = \frac{\widetilde{X}}{\sigma^2(A)}. \tag{13.1}$$

At $\beta_0$, $Y_{\beta_0}^\circ = Y - \beta_0 X = \mu_0(A) + U$, and $M_0^P \mu_0 = 0$. Hence

$$M_0^P Y_{\beta_0}^\circ \, M_0^P w^* = U\frac{\widetilde{X}}{\sigma^2(A)} = S_{\text{eff}}.$$

This is the population reason the profile score becomes efficient when $w = w^*$. If $\sigma^2(A)$ is constant, multiplying the moment by the constant $1/\sigma^2$ does not change the root, so $w = X$ is the efficient choice in the homoskedastic case.

For the sample statement, define

$$w_t^* = \frac{x_t}{\sigma^2(A_t)}, \qquad \boldsymbol{w}^* = (w_1^*, \dots, w_T^*)', \qquad \boldsymbol{v}_{K,T}^* = \boldsymbol{M}_0 \boldsymbol{w}^*,$$

where $A_t = (J_{0,t}, W_{t-1})$. Here $J_{0,t} = j$ iff $k_{j,0} < t \le k_{j+1,0}$, and $\boldsymbol{M}_0 = \boldsymbol{M}_{K,\Lambda_0}$. The efficient sieve approximation target is

$$\left\| \boldsymbol{v}_{K,T}^* - \left( \frac{\widetilde{X}_1}{\sigma^2(A_1)}, \dots, \frac{\widetilde{X}_T}{\sigma^2(A_T)} \right)' \right\|_T = o_p(1). \tag{13.2}$$

This follows from the same finite-sieve approximation and Gram-stability logic used in Part II, provided the block sieve approximates $E(w^* \mid A) = E(X \mid A)/\sigma^2(A)$.

**Lemma 20** (Efficient-score equivalence)**.** *Assume* (13.2) *and the true-partition efficient-weight bias condition*

$$T^{-1/2} \boldsymbol{r}_{\mu,T}' \boldsymbol{v}_{K,T}^* = o_p(1), \qquad \boldsymbol{r}_{\mu,T} = \boldsymbol{M}_0 \boldsymbol{\mu}_0.$$

*Then*

$$\sqrt{T}\Psi_T^*(\beta_0, \Lambda_0) = T^{-1/2} \sum_{t=1}^{T} S_{\text{eff},t} + o_p(1),$$

*where*

$$\Psi_T^*(\beta, \Lambda) = \langle \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y} - \beta \boldsymbol{x}), \boldsymbol{M}_{K,\Lambda} \boldsymbol{w}^* \rangle_T.$$

*Proof.* At $\beta_0$,

$$\boldsymbol{y}_{\beta_0}^{\circ} = \boldsymbol{\mu}_0 + \boldsymbol{u}.$$

Therefore

$$\begin{aligned} \sqrt{T}\Psi_T^*(\beta_0, \Lambda_0) &= T^{-1/2} (\boldsymbol{M}_0 \boldsymbol{y}_{\beta_0}^{\circ})' \boldsymbol{v}_{K,T}^* \\ &= T^{-1/2} \boldsymbol{u}' \boldsymbol{v}_{K,T}^* + T^{-1/2} \boldsymbol{r}_{\mu,T}' \boldsymbol{v}_{K,T}^*. \end{aligned}$$

The second term is $o_p(1)$ by assumption. The first term equals

$$\begin{aligned} T^{-1/2} \boldsymbol{u}' \boldsymbol{v}_{K,T}^* &= T^{-1/2} \sum_{t=1}^{T} U_t \frac{\widetilde{X}_t}{\sigma^2(A_t)} + o_p(1) \\ &= T^{-1/2} \sum_{t=1}^{T} S_{\text{eff},t} + o_p(1). \end{aligned}$$

by (13.2) and the same Cauchy-Schwarz/variance argument used for the score replacement in Part II. $\square$

For fixed $\Lambda$, the efficient-weight moment is linear in $\beta$, with

$$\frac{\partial}{\partial \beta}\Psi_T^*(\beta, \Lambda) = -\langle \boldsymbol{M}_{K,\Lambda}\boldsymbol{x}, \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}^* \rangle_T.$$

Thus the derivative denominator in the root expansion is the empirical version of the efficient information.

**Theorem 1** (Optional semiparametric efficiency with efficient weight)**.** *Let $\widehat{\beta}_T$ be a profile EL root, or an approximate profile moment root, satisfying*

$$\Psi_T^*(\widehat{\beta}_T, \widehat{\Lambda}) = o_p(T^{-1/2}).$$

*Assume the efficient-score equivalence in Lemma 20, estimated-partition score equivalence at the $T^{-1/2}$ scale, and derivative equivalence:*

$$\sqrt{T}\{\Psi_T^*(\beta_0, \widehat{\Lambda}) - \Psi_T^*(\beta_0, \Lambda_0)\} = o_p(1),$$

$$\langle \boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{x}, \boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{w}^* \rangle_T - \langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{v}_{K,T}^* \rangle_T = o_p(1).$$

*Assume also*

$$\langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{v}_{K,T}^* \rangle_T \to_p I_{\text{eff}} > 0.$$

*Then*

$$\sqrt{T}(\widehat{\beta}_T - \beta_0) = I_{\text{eff}}^{-1} T^{-1/2} \sum_{t=1}^{T} S_{\text{eff},t} + o_p(1),$$

*and consequently*

$$\sqrt{T}(\widehat{\beta}_T - \beta_0) \Rightarrow N(0, I_{\text{eff}}^{-1}).$$

*Thus the efficient-weight profile-sieve EL root attains the semiparametric efficiency bound.*

*Proof.* By a mean-value expansion of the projected moment around $\beta_0$,

$$\begin{aligned} 0 &= \Psi_T^*(\widehat{\beta}_T, \widehat{\Lambda}) + o_p(T^{-1/2}) \\ &= \Psi_T^*(\beta_0, \widehat{\Lambda}) - (\widehat{\beta}_T - \beta_0)\langle \boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{x}, \boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{w}^* \rangle_T + o_p(T^{-1/2}) + o_p(|\widehat{\beta}_T - \beta_0|). \end{aligned}$$

The score-equivalence and derivative-equivalence assumptions replace the estimated partition by the true partition. Hence

$$\sqrt{T}(\widehat{\beta}_T - \beta_0) = \left\{\langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{v}_{K,T}^* \rangle_T\right\}^{-1} \sqrt{T}\Psi_T^*(\beta_0, \Lambda_0) + o_p(1).$$

The denominator converges to

$$E\{(M_0^P X)(M_0^P w^*)\} = E\left\{\widetilde{X}\frac{\widetilde{X}}{\sigma^2(A)}\right\} = I_{\text{eff}}.$$

Lemma 20 gives

$$\sqrt{T}\Psi_T^*(\beta_0, \Lambda_0) = T^{-1/2}\sum_{t=1}^{T} S_{\text{eff},t} + o_p(1).$$

This proves the asymptotic linear representation. Since $E(S_{\text{eff}} \mid X, A) = 0$ and $E(S_{\text{eff}}^2) = I_{\text{eff}}$, the relevant martingale or iid CLT gives

$$T^{-1/2} \sum_{t=1}^{T} S_{\text{eff},t} \Rightarrow N(0, I_{\text{eff}}).$$

Therefore

$$\sqrt{T}(\widehat{\beta}_T - \beta_0) \Rightarrow N(0, I_{\text{eff}}^{-1}).$$

The standard Cauchy-Schwarz information bound gives $E(\phi^2) \geq I_{\text{eff}}^{-1}$ for any regular influence function $\phi$ satisfying $E\{\phi S_{\text{eff}}\} = 1$. The influence function above is $I_{\text{eff}}^{-1} S_{\text{eff}}$, whose variance is $I_{\text{eff}}^{-1}$. It therefore attains the bound. □

**Remark 1** (Estimated variance plug-in)**.** The main theorem is cleanest with known $\sigma^2(A)$, or with the homoskedastic simplification $w = X$. If $\sigma^2(A)$ is estimated, define

$$\widehat{w}_t^* = \frac{x_t}{\widehat{\sigma}^2(A_t)}, \qquad \widehat{\boldsymbol{w}}^* = (\widehat{w}_1^*, \ldots, \widehat{w}_T^*)'.$$

Replacing $\boldsymbol{w}^*$ by $\widehat{\boldsymbol{w}}^*$ leaves the same asymptotic linear representation if

$$\|\boldsymbol{M}_{K,\widehat{\Lambda}}(\widehat{\boldsymbol{w}}^* - \boldsymbol{w}^*)\|_T = o_p(1),$$

$$\sqrt{T}\left|\langle \boldsymbol{M}_{K,\widehat{\Lambda}}(\boldsymbol{y} - \beta_0 \boldsymbol{x}), \boldsymbol{M}_{K,\widehat{\Lambda}}(\widehat{\boldsymbol{w}}^* - \boldsymbol{w}^*)\rangle_T\right| = o_p(1),$$

and

$$\left|\langle \boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{x}, \boldsymbol{M}_{K,\widehat{\Lambda}}(\widehat{\boldsymbol{w}}^* - \boldsymbol{w}^*)\rangle_T\right| = o_p(1).$$

This plug-in statement is an appendix-level add-on; the main optional theorem should use either $w = X$ under homoskedasticity or the oracle heteroskedastic weight $w^* = X/\sigma^2(A)$.

This theorem is conditional on the efficiency model and the efficient-weight choice. With only martingale exogeneity and no conditional variance or local likelihood structure, the workbook proves valid profile EL Wilks inference, but not a Fisher-style semiparametric efficiency bound.

# 14 Three-Part Logic Checklist

| Proof part | Object | What it proves |
|---|---|---|
| Geometry / KKT | $\boldsymbol{y} = \beta_0 \boldsymbol{x} + \boldsymbol{\mu}_0 + \boldsymbol{u} \in L^2(P_T)$ | The full sample model is a Hilbert-space identity. |
| Geometry / KKT | $\mathcal{N}_{K,\Lambda,T} = \mathrm{col}(Q_{K,\Lambda})$ | The sieve basis represents the nuisance tangent plane. |
| Geometry / KKT | $M_{K,\Lambda} = I - \Pi_{K,\Lambda,T}$ | Profiling is the KKT normal equation for orthogonal projection. |
| Geometry / KKT | $g_t = \{M(y - \beta x)\}_t \{Mw\}_t$ | The profile score is one projected inner product. |
| Geometry / KKT | EL primal/dual KKT | EL is a convex simplex problem cut by the moment hyperplane. |
| Population geometry | $\Psi(\beta, \Lambda_0) = -(\beta - \beta_0)\Gamma$ | Exogeneity and relevance give the unique population zero. |
| Small stochastic | $\sup_\beta \lvert\Psi_T - \Psi\rvert = o_p(1)$ | The sample projected moment reaches the population moment. |
| Small stochastic | CLT, variance, max-score, convex hull | The EL KKT expansion is valid locally and the standardized score is normal. |
| Merging | Moment bridge plus unique zero | Consistency: $\widehat{\beta}_T \to_p \beta_0$. |
| Merging | EL KKT quadratic reduction plus CLT | Wilks: $\ell_T(\beta_0, \widehat{\Lambda}) \Rightarrow \chi^2_1$. |

# 15 Minimal Final Statement

The full profile-sieve EL proof is a three-part argument.

**Geometric/KKT body.** In $L^2(P_T)$, the model is $\boldsymbol{y} = \beta_0 \boldsymbol{x} + \boldsymbol{\mu}_0 + \boldsymbol{u}$. A candidate partition and sieve define the nuisance tangent plane $\mathcal{N}_{K,\Lambda,T} = \mathrm{col}(Q_{K,\Lambda})$. Profiling is orthogonal projection onto this space. The residual score is the projected inner product

$$\Psi_T(\beta, \Lambda) = \langle M_{K,\Lambda}(\boldsymbol{y} - \beta \boldsymbol{x}), M_{K,\Lambda} \boldsymbol{w} \rangle_T.$$

EL is then a convex simplex problem with the moment hyperplane $\sum_t \pi_t g_t = 0$, whose KKT system gives $\pi_t = 1/[T\{1 + \lambda g_t\}]$.

**Small stochastic body.** The stochastic lemmas verify that the empirical projection, estimated break partition, sieve approximation, score variance, convex hull, and max-score terms are small enough for the sample KKT objects to approximate their population limits.

**Merging body.** For consistency, combine $\Psi_T \to \Psi$ with the unique population zero. For Wilks, combine the EL KKT quadratic reduction with the standardized score CLT:

$$\ell_T(\beta_0, \widehat{\Lambda}) = \frac{T\Psi_T(\beta_0, \widehat{\Lambda})^2}{V_T(\widehat{\Lambda})} + o_p(1), \qquad \frac{\sqrt{T}\Psi_T(\beta_0, \widehat{\Lambda})}{\sqrt{V_T(\widehat{\Lambda})}} \Rightarrow N(0, 1).$$

Therefore $\ell_T(\beta_0, \widehat{\Lambda}) \Rightarrow \chi_1^2$.